

Interpolation and Polynomial Approximation

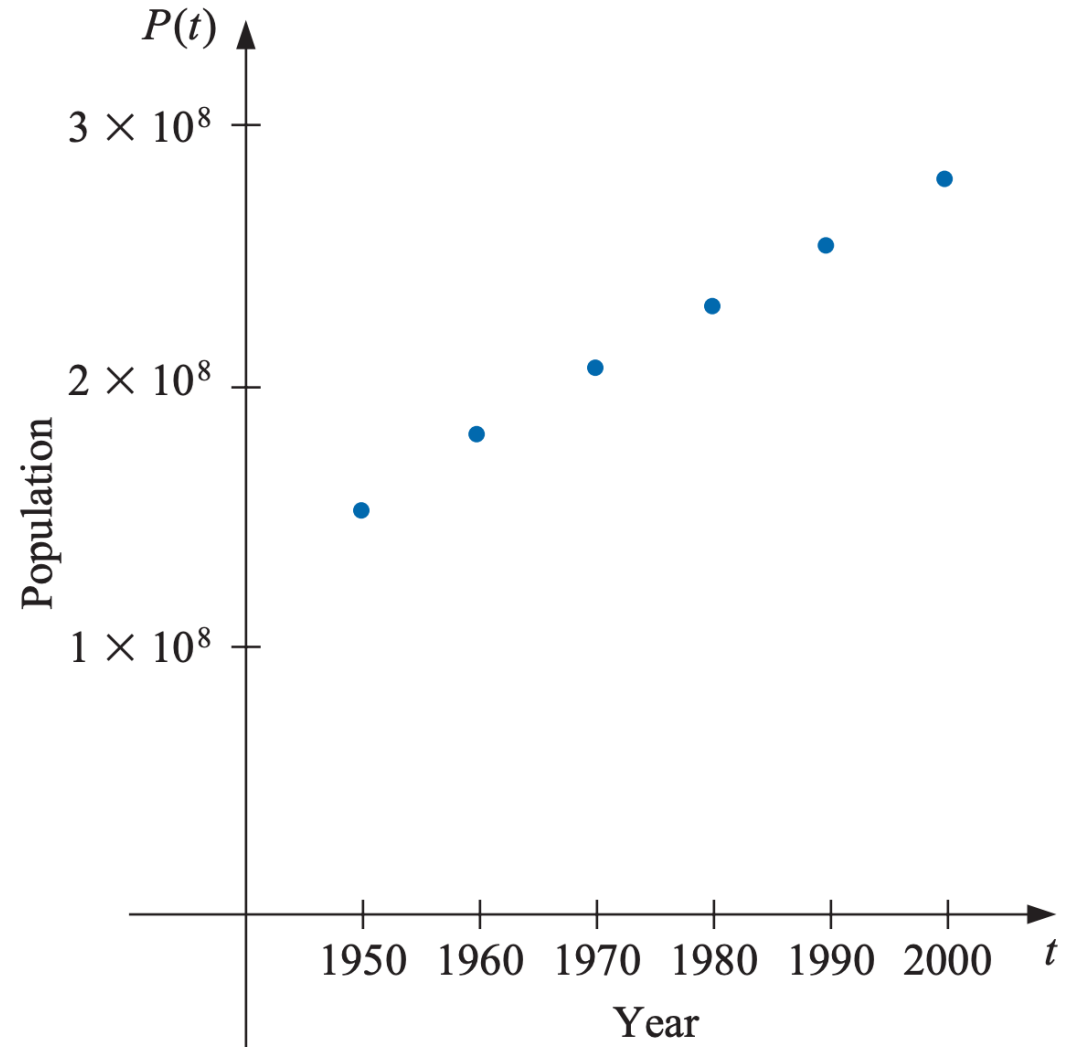
NUMERICAL ANALYSIS - APRIL, 2026

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Introduction

- ❖ A census of the population of a country is taken in 1998 and the following table lists the population, in lakhs, for each state. In 2000, and the data are also represented by a bar graph.

Year	1950	1960	1970
Population (in thousands)	151,326	179,323	203,312



Introduction

- ❖ A census of the population of a country is taken every 10 years. The following table lists the population, in thousands of people, from 1950 to 2000.

Year	1950	1960	1970	1980	1990	2000
Population (in thousands)	151,326	179,323	203,302	226,542	249,633	281,422

- ❖ In reviewing these data, we might ask whether they could be used to provide a reasonable estimate of the population, say, in 1975 or even in the year 2026.
- ❖ Predictions of this type can be obtained by using a function that fits the given data. This process is called *interpolation*.

Interpolation

❖ One of the most useful and well-known classes of functions mapping the set of real numbers into itself is the algebraic polynomials.

✓ The set of functions of the form

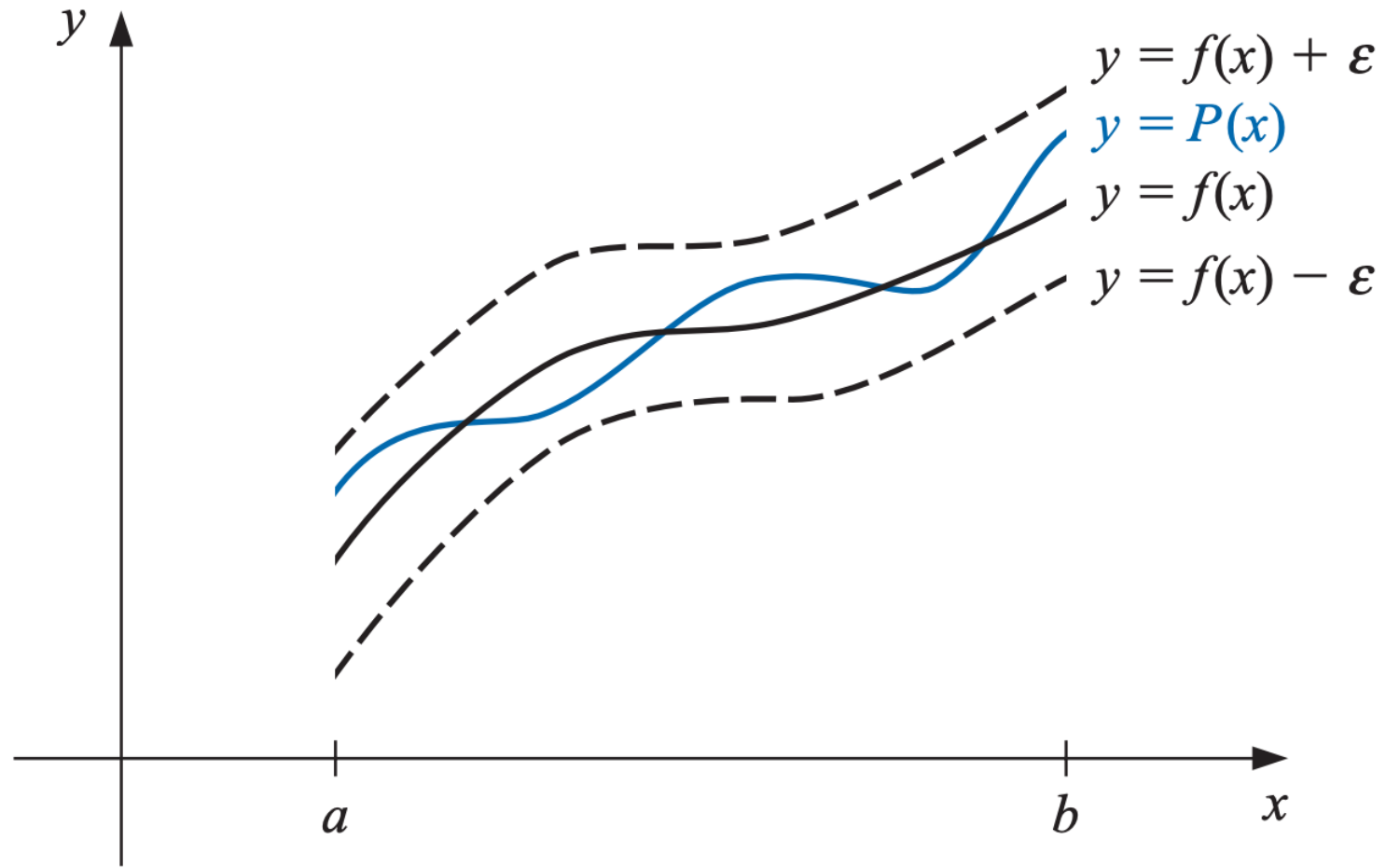
$$P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$$

where n is a nonnegative integer and $a_0, \dots, a_n$ are real constants.

❖ One reason for their importance is that they uniformly approximate continuous functions.

✓ i.e. Given any function, defined and continuous on a closed and bounded interval, there exists a polynomial that is as “close” to the given function as desired.

Interpolation



Interpolation

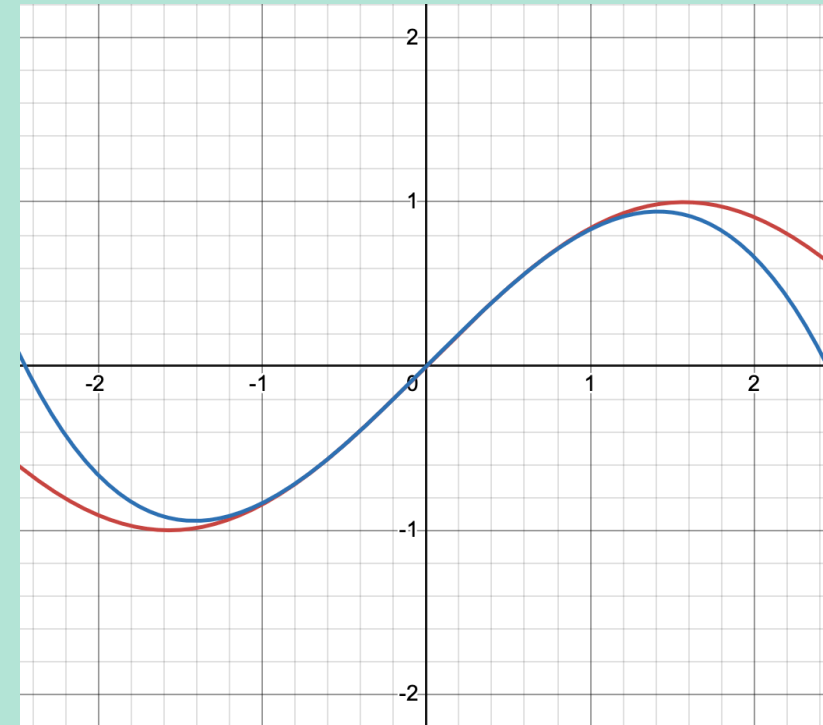
Theorem: Weierstrass Approximation Theorem.

Suppose that f is defined and continuous on $[a, b]$. For each $\varepsilon > 0$, there exists a polynomial $P(x)$, with the property that $|f(x) - P(x)| < \varepsilon$, for all x in $[a, b]$.

Example: Use graphing calculator to approximate $f(x) = \sin x$

On the interval $[-1, 1]$, the function $f(x) = \sin x$ can be approximated using the function

$$P(x) = x - \frac{x^3}{6}$$



Interpolation

- ❖ Another important reason for considering the class of polynomials in the approximation of functions is that the derivative and indefinite integral of a polynomial are easy to determine and are also polynomials.
- ❖ A good interpolation polynomial needs to provide a relatively accurate approximation over an entire interval.
 - ✓ we do not simply approximate values at a point.
 - ✓ we approximate the “behavior” of a function on an interval.

Lagrange Interpolating Polynomials

- ❖ Suppose we have two points (x_0, y_0) and (x_1, y_1)
 - ✓ Recall that if we have a function f with two points (x_0, y_0) and (x_1, y_1) , then we have

$$\begin{aligned}f(x_0) &= y_0 \\f(x_1) &= y_1\end{aligned}$$

- ✓ determining a polynomial of degree one that passes through these distinct points is the same as approximating a function f for which $f(x_0) = y_0$ and $f(x_1) = y_1$ by means of a first-degree polynomial interpolating, or agreeing with, the values of f at the given points.
- ❖ Using this polynomial for approximation within the interval given by the endpoints is called polynomial interpolation.

Lagrange Interpolating Polynomials

❖ We define

$$L_0(x) = \frac{x - x_1}{x_0 - x_1} \quad \text{and} \quad L_1(x) = \frac{x - x_0}{x_1 - x_0}$$

The linear Lagrange interpolating polynomial through (x_0, y_0) and (x_1, y_1) is

$$P(x) = L_0(x)f(x_0) + L_1(x)f(x_1)$$

$$P(x) = \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1)$$

Lagrange Interpolating Polynomials

Observe that

$$L_0(x_0) = \frac{x_0 - x_1}{x_0 - x_1} = 1 \quad \text{and} \quad L_1(x_0) = \frac{x_0 - x_0}{x_1 - x_0} = 0$$

$$L_0(x_1) = \frac{x_1 - x_1}{x_0 - x_1} = 0 \quad \text{and} \quad L_1(x_1) = \frac{x_1 - x_0}{x_1 - x_0} = 1$$

Hence, we have

$$P(x_0) = L_0(x_0)f(x_0) + L_1(x_0)f(x_1) = (1)f(x_0) + (0)f(x_1) = f(x_0) = y_0$$

$$P(x_1) = L_0(x_1)f(x_0) + L_1(x_1)f(x_1) = (0)f(x_0) + (1)f(x_1) = f(x_1) = y_1$$

So P is the unique polynomial of degree at most one that passes through (x_0, y_0) and (x_1, y_1) .

Lagrange Interpolating Polynomials

Example: Determine the linear Lagrange interpolating polynomial that passes through the points (2, 4) and (5, 1).

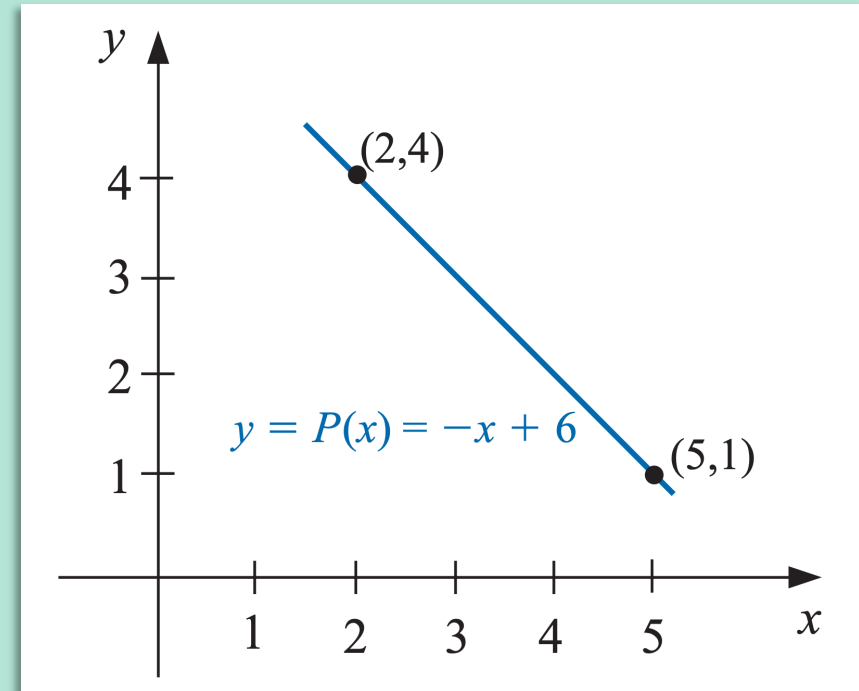
Solution: Given points (2, 4) and (5, 1), we let $x_0 = 2$ and $x_1 = 5$. It follows that $f(x_0) = 4$ while $f(x_1) = 1$. Thus we have

$$P(x) = \frac{x - x_1}{x_0 - x_1} f(x_0) + \frac{x - x_0}{x_1 - x_0} f(x_1)$$

$$P(x) = \frac{x - (5)}{(2) - (5)} (4) + \frac{x - (2)}{(5) - (2)} (1)$$

$$P(x) = \frac{4x - 20}{-3} + \frac{x - 2}{3}$$

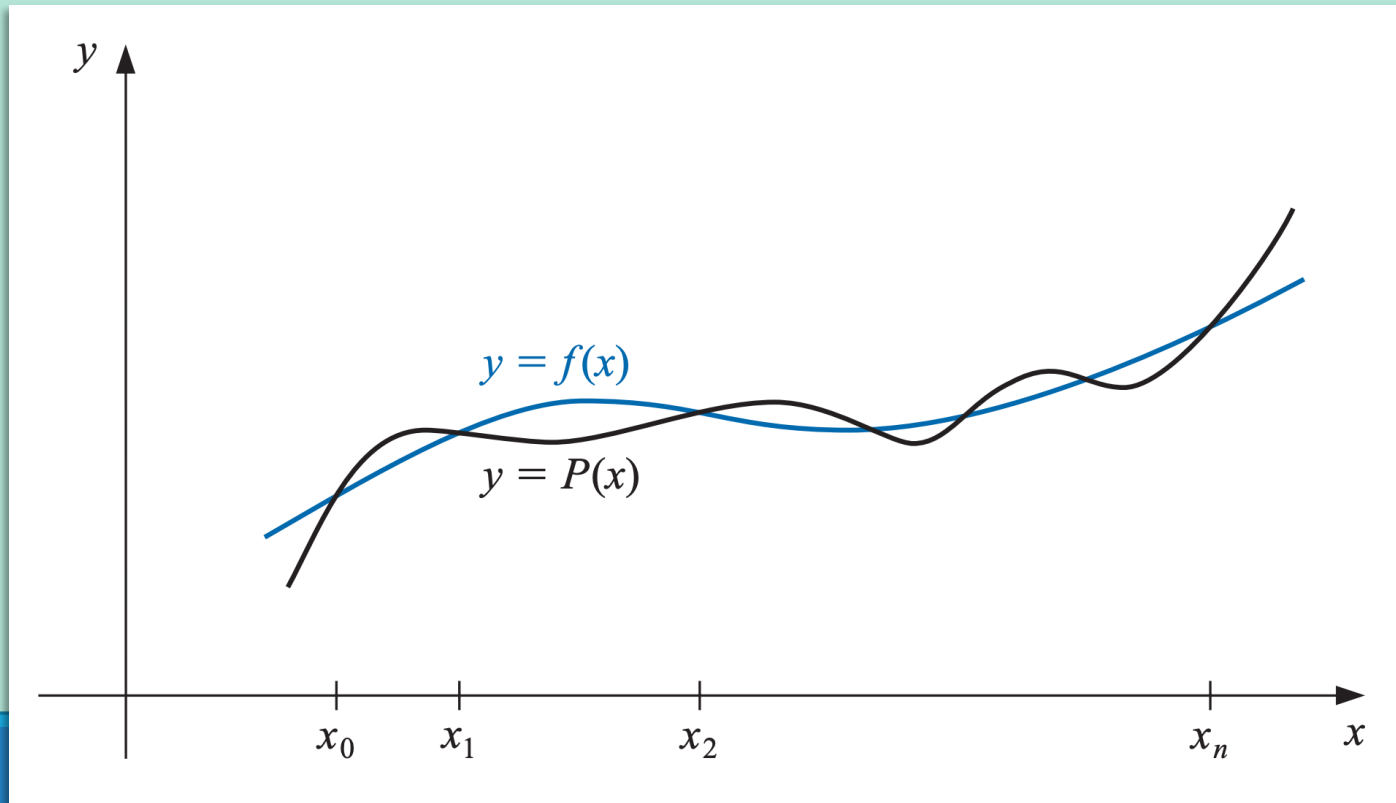
$$P(x) = -x + 6 \quad \text{Answer.}$$



Lagrange Interpolating Polynomials

- ❖ We generalize the concept of linear interpolation, extending it to the construction of a polynomial of degree at most n that passes through the $n + 1$ points

$$(x_0, f(x_0)), (x_1, f(x_1)), (x_2, f(x_2)), \dots, (x_n, f(x_n))$$



Lagrange Interpolating Polynomials

- ❖ We generalize the concept of linear interpolation, extending it to the construction of a polynomial of degree at most n that passes through the $n + 1$ points

$$(x_0, f(x_0)), (x_1, f(x_1)), (x_2, f(x_2)), \dots, (x_n, f(x_n))$$

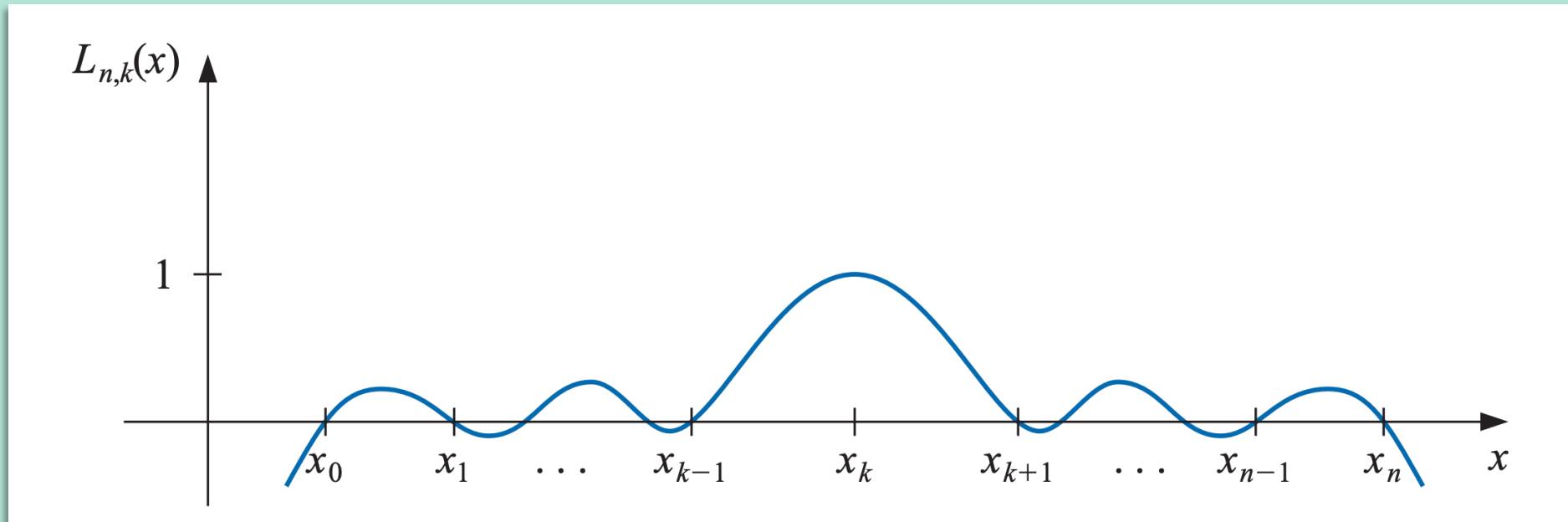
- ❖ In this case, we first construct a function $L_{n,k}(x)$ with the property that $L_{n,k}(x_i) = 0$ when $i \neq k$ and $L_{n,k}(x_k) = 1$, for each $k = 0, 1, \dots, n$.
 - ✓ To satisfy $L_{n,k}(x_i) = 0$ for each $i \neq k$ requires that the numerator of $L_{n,k}(x)$ contain the term
$$(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n).$$
 - ✓ To satisfy $L_{n,k}(x_k) = 1$, the denominator must be of the form
$$(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n).$$

Lagrange Interpolating Polynomials

❖ Hence, we have

$$L_{n,k}(x) = \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}$$

❖ A sketch of the graph of a typical $L_{n,k}$ (when n is even) is given by



Lagrange Interpolating Polynomials

Theorem: If $x_0, x_1, \dots, x_n$ are $n + 1$ distinct numbers and f is a function whose values are given at these numbers, then a unique polynomial $P(x)$ of degree at most n exists with $f(x_k) = P(x_k)$, for each $k = 0, 1, \dots, n$

This polynomial is given by

$$P(x) = f(x_0)L_{n,0}(x) + \dots + f(x_n)L_{n,n}(x) = \sum_{k=0}^n f(x_k)L_{n,k}(x)$$

where, for each $k = 0, 1, \dots, n$,

$$L_{n,k}(x) = \frac{(x - x_0)(x - x_1) \cdots (x - x_{k-1})(x - x_{k+1}) \cdots (x - x_n)}{(x_k - x_0)(x_k - x_1) \cdots (x_k - x_{k-1})(x_k - x_{k+1}) \cdots (x_k - x_n)}$$
$$= \prod_{\substack{i=0 \\ i \neq k}}^n \frac{(x - x_i)}{(x_k - x_i)}$$

We will write $L_{n,k}(x)$ simply as $L_k(x)$ when there is no confusion as to its degree.

Lagrange Interpolating Polynomials

Example: (a) Use the numbers (called nodes) $x_0 = 2$, $x_1 = 2.75$, and $x_2 = 4$ to find the second Lagrange interpolating polynomial for $f(x) = \frac{1}{x}$.

(b) Use this polynomial to approximate $f(3) = \frac{1}{3}$.

Answer: (a) $P(x) = \frac{1}{22}x^2 - \frac{35}{88}x + \frac{49}{44}$

(b) $P(3) = \frac{29}{88} \approx 0.32955$

3 x-values

$$f(x) = \frac{1}{x}$$

NOTE: n number of
x-values then
we get

Lagrange polynomial
of order $n-1$

Example: In the previous example, we found the second Lagrange polynomial for $f(x) = \frac{1}{x}$ on $[2, 4]$ using the nodes $x_0 = 2$, $x_1 = 2.75$, and $x_2 = 4$. Determine the error form for this polynomial, and the maximum error when the polynomial is used to approximate $f(x)$ for $x \in [2, 4]$.

Sol'n: $n=2$

$$L_{2,k}(x) \begin{cases} L_{2,0}(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} \\ L_{2,1}(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} \\ L_{2,2}(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} \end{cases}$$

$$f(x_0) = f(2) = \frac{1}{2}$$

$$f(x_1) = f(2.75) = f\left(\frac{11}{4}\right) = \frac{1}{\left(\frac{11}{4}\right)} = \frac{4}{11}$$

$$f(x_2) = f(4) = \frac{1}{4}$$

$$p(x) = \left(\frac{1}{2}\right) \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} + \frac{4}{11} \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + \frac{1}{4} \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

Note

$$L_{2,k}(x) = \frac{(x-x_0)(x-x_1)(x-x_2)}{(x_k-x_0)(x_k-x_1)(x_k-x_2)}$$

we remove the factors leading to zero in the numerator, and on the denominator

$$2 + \frac{3}{4}$$

Given $x_0 = 2$, $x_1 = 2.75 = \frac{11}{4}$ and $x_2 = 4$, then

$$\begin{aligned}
 P(x) &= \left(\frac{1}{2}\right) \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_1-x_2)} + \left(\frac{\frac{4}{11}}{11}\right) \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} + \left(\frac{1}{4}\right) \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} \\
 &= \left(\frac{1}{2}\right) \frac{(x-\frac{11}{4})(x-4)}{(2-\frac{11}{4})(2-4)} + \left(\frac{\frac{4}{11}}{11}\right) \frac{(x-2)(x-4)}{(\frac{11}{4}-2)(\frac{11}{4}-4)} + \left(\frac{1}{4}\right) \frac{(x-2)(x-\frac{11}{4})}{(4-2)(4-\frac{11}{4})} \\
 &= \left(\frac{1}{2}\right) \frac{(\frac{1}{4})(4x-11)(x-4)}{(\frac{8}{4}-\frac{11}{4})(-2)} + \left(\frac{\frac{4}{11}}{11}\right) \frac{(x-2)(x-4)}{(\frac{11}{4}-\frac{8}{4})(\frac{11}{4}-\frac{16}{4})} + \frac{1}{4} \frac{(x-2)(\frac{1}{4})(4x-11)}{(2)(\frac{16}{4}-\frac{11}{4})} \\
 &= \left(\frac{1}{2}\right) \frac{(\frac{1}{4})(4x-11)(x-4)}{(-\frac{3}{4})(-2)} + \left(\frac{\frac{4}{11}}{11}\right) \frac{(x-2)(x-4)}{(\frac{3}{4})(-\frac{5}{4})} + \frac{1}{4} \frac{(\frac{1}{4})(x-2)(4x-11)}{(2)(\frac{5}{4})} \\
 &= \left(\frac{1}{4}\right) \left(\frac{1}{6}\right) (4x-11)(x-4) + \frac{4}{11} \left(-\frac{16}{15}\right) (x-2)(x-4) + \left(\frac{1}{4}\right) \left(\frac{1}{10}\right) (x-2)(4x-11)
 \end{aligned}$$

$2(-\frac{3}{4})(-2) = 3$
 $(\frac{3}{4})(-\frac{5}{4}) = -\frac{15}{16}$
 $(4)(2)(\frac{5}{4}) = 10$

Lagrange Interpolating Polynomials

Theorem: Suppose $x_0, x_1, \dots, x_n$ are $n + 1$ distinct numbers in the interval $[a, b]$ and $f \in C^{n+1}[a, b]$. Then, for each x in $[a, b]$, a number $\xi(x)$ (generally unknown) between $x_0, x_1, \dots, x_n$, and hence in (a, b) , exists with

$$f(x) = P(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n)$$

where $P(x)$ is the interpolating polynomial.

- ✓ The error formula in this theorem is an important theoretical result because Lagrange polynomials are used extensively for deriving numerical differentiation and integration methods.
- ✓ Error bounds for these techniques are obtained from the Lagrange error formula

Lagrange Interpolating Polynomials

Example: In the previous example, we found the second Lagrange polynomial for $f(x) = \frac{1}{x}$ on $[2, 4]$ using the nodes $x_0 = 2$, $x_1 = 2.75$, and $x_2 = 4$.

Determine the error form for this polynomial, and the maximum error when the polynomial is used to approximate $f(x)$ for $x \in [2, 4]$.

Sol'n: $n = 2$

$$n+1 = 3$$

$$(n+1)! = (3)(2)(1) = 6$$

Hence, given $f(x) = \frac{1}{x} \rightarrow f'(x) = \frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = -x^{-2}$

$$f''(x) = \frac{d}{dx}(-x^{-2}) = (-1)(-2)x^{-3} = 2x^{-3}$$

$$f'''(x) = -6x^{-4} \quad \leftarrow \text{replace all } x \text{ by } \xi(x)$$

Let E_f be the error form.

$$E_f = \frac{-6(\xi(x))^{-4}}{3!} (x-2)\left(x-\frac{11}{4}\right)(x-4)$$

$$= -\frac{1}{(\xi(x))^4} (x-2)\left(x-\frac{11}{4}\right)(x-4)$$

Example: In the previous example, we found the second Lagrange polynomial for $f(x) = \frac{1}{x}$ on $[2, 4]$ using the nodes $x_0 = 2$, $x_1 = 2.75$, and $x_2 = 4$. Determine the error form for this polynomial, and the maximum error when the polynomial is used to approximate $f(x)$ for $x \in [2, 4]$.

We want to determine the max error, the same as

$$\max \left| -\frac{1}{(f(x))^4} (x-2)(x-\frac{11}{4})(x-4) \right| \rightarrow \max \left| -\frac{1}{(f(x))^4} \right|$$

on the interval $[2, 4]$

$$\max |(x-2)(x-\frac{11}{4})(x-4)|$$

For $\max \left| -\frac{1}{(f(x))^4} \right| < \begin{matrix} -\frac{1}{2^4} = -\frac{1}{16} \\ -\frac{1}{4^4} = -\frac{1}{256} \end{matrix} \rightarrow \frac{1}{16} \text{ vs } \frac{1}{256}$

$$\max \left| -\frac{1}{(f(x))^4} \right| = \frac{1}{16}$$

Let $g(x) = (x-2)(x-\frac{11}{4})(x-4) = x^3 - \frac{35}{4}x^2 + \frac{49}{2}x - 22$

$$g'(x) = 3x^2 - \frac{35}{2}x + \frac{49}{2} = \frac{1}{2}(6x^2 - 35x + 49) = \frac{1}{2}(3x-7)(2x-7)$$

Hence $g'(x) = 0$ when $x = \frac{7}{3}$ and $x = \frac{7}{2}$.

$3x-7=0 \Rightarrow x = \frac{7}{3}$
 $2x-7=0 \Rightarrow x = \frac{7}{2}$

Example: In the previous example, we found the second Lagrange polynomial for $f(x) = \frac{1}{x}$ on $[2, 4]$ using the nodes $x_0 = 2$, $x_1 = 2.75$, and $x_2 = 4$. Determine the error form for this polynomial, and the maximum error when the polynomial is used to approximate $f(x)$ for $x \in [2, 4]$.

we evaluate $g(x) = (x-2)(x-\frac{11}{4})(x-4)$

at $x=2$ $g(2) = (2-2)(2-\frac{11}{4})(2-4) = 0$

at $x=4$ $g(4) = (4-2)(4-\frac{11}{4})(4-4) = 0$

at critical #s $x = \frac{7}{3}$ $g(\frac{7}{3}) = (\frac{7}{3}-2)(\frac{7}{3}-\frac{11}{4})(\frac{7}{3}-4) = \frac{25}{108}$

$x = \frac{7}{2}$, $g(\frac{7}{2}) = (\frac{7}{2}-2)(\frac{7}{2}-\frac{11}{4})(\frac{7}{2}-4) = -\frac{9}{16}$

$\max |g(x)| = \max |(x-2)(x-\frac{11}{4})(x-4)| = \frac{9}{16}$

hence we have

$$\left| -\frac{1}{f(x)^3} \right| |(x-2)(x-\frac{11}{4})(x-4)| \leq \left(\frac{1}{16} \right) \left(\frac{9}{16} \right) = \left(\frac{9}{256} \right)$$

$$\begin{array}{r} 400 \qquad 972 \\ \hline 25 \qquad 9 \\ \hline 108 \qquad 16 \end{array}$$

$$\begin{array}{r} 3 \text{ } 16 \\ \hline 25 \\ \hline 80 \\ \hline 32 \end{array}$$

Example: In the previous example, we found the second Lagrange polynomial for $f(x) = \frac{1}{x}$ on $[2, 4]$ using the nodes $x_0 = 2$, $x_1 = 2.75$, and $x_2 = 4$. Determine the error form for this polynomial, and the maximum error when the polynomial is used to approximate $f(x)$ for $x \in [2, 4]$.

Solution Because $f(x) = x^{-1}$, we have

$$f'(x) = -x^{-2}, \quad f''(x) = 2x^{-3}, \quad \text{and} \quad f'''(x) = -6x^{-4}.$$

As a consequence, the second Lagrange polynomial has the error form

$$\frac{f'''(\xi(x))}{3!}(x-x_0)(x-x_1)(x-x_2) = -(\xi(x))^{-4}(x-2)(x-2.75)(x-4), \quad \text{for } \xi(x) \text{ in } (2, 4).$$

The maximum value of $(\xi(x))^{-4}$ on the interval is $2^{-4} = 1/16$. We now need to determine the maximum value on this interval of the absolute value of the polynomial

$$g(x) = (x-2)(x-2.75)(x-4) = x^3 - \frac{35}{4}x^2 + \frac{49}{2}x - 22.$$

Because

$$D_x \left(x^3 - \frac{35}{4}x^2 + \frac{49}{2}x - 22 \right) = 3x^2 - \frac{35}{2}x + \frac{49}{2} = \frac{1}{2}(3x-7)(2x-7),$$

the critical points occur at

$$x = \frac{7}{3}, \text{ with } g\left(\frac{7}{3}\right) = \frac{25}{108}, \quad \text{and} \quad x = \frac{7}{2}, \text{ with } g\left(\frac{7}{2}\right) = -\frac{9}{16}.$$

Hence, the maximum error is

$$\frac{f'''(\xi(x))}{3!} |(x-x_0)(x-x_1)(x-x_2)| \leq \frac{1}{16 \cdot 6} \left| -\frac{9}{16} \right| = \frac{3}{512} \approx 0.00586. \quad \blacksquare$$

Lagrange Interpolating Polynomials

Example: Suppose a table is to be prepared for the function $f(x) = e^x$, for x in $[0, 1]$. Assume the number of decimal places to be given per entry is $d \geq 8$ and that the difference between adjacent x -values, the step size, is h . What step size h will ensure that linear interpolation gives an absolute error of at most 10^{-6} for all x in $[0, 1]$.

Solution Let $x_0, x_1, \dots$ be the numbers at which f is evaluated, x be in $[0,1]$, and suppose j satisfies $x_j \leq x \leq x_{j+1}$. Eq. (3.3) implies that the error in linear interpolation is

$$|f(x) - P(x)| = \left| \frac{f^{(2)}(\xi)}{2!} (x - x_j)(x - x_{j+1}) \right| = \frac{|f^{(2)}(\xi)|}{2} |(x - x_j)|(x - x_{j+1})|.$$

The step size is h , so $x_j = jh$, $x_{j+1} = (j+1)h$, and

$$|f(x) - P(x)| \leq \frac{|f^{(2)}(\xi)|}{2!} |(x - jh)(x - (j+1)h)|.$$

Hence

$$\begin{aligned} |f(x) - P(x)| &\leq \frac{\max_{\xi \in [0,1]} e^{\xi}}{2} \max_{x_j \leq x \leq x_{j+1}} |(x - jh)(x - (j+1)h)| \\ &\leq \frac{e}{2} \max_{x_j \leq x \leq x_{j+1}} |(x - jh)(x - (j+1)h)|. \end{aligned}$$

Consider the function $g(x) = (x - jh)(x - (j+1)h)$, for $jh \leq x \leq (j+1)h$. Because

$$g'(x) = (x - (j+1)h) + (x - jh) = 2 \left(x - jh - \frac{h}{2} \right),$$

the only critical point for g is at $x = jh + h/2$, with $g(jh + h/2) = (h/2)^2 = h^2/4$.

Lagrange Interpolating Polynomials

Since $g(jh) = 0$ and $g((j+1)h) = 0$, the maximum value of $|g'(x)|$ in $[jh, (j+1)h]$ must occur at the critical point which implies that

$$|f(x) - P(x)| \leq \frac{e}{2} \max_{x_j \leq x \leq x_{j+1}} |g(x)| \leq \frac{e}{2} \cdot \frac{h^2}{4} = \frac{eh^2}{8}.$$

Consequently, to ensure that the error in linear interpolation is bounded by 10^{-6} , it is sufficient for h to be chosen so that

$$\frac{eh^2}{8} \leq 10^{-6}. \quad \text{This implies that } h < 1.72 \times 10^{-3}.$$

Because $\frac{n}{1000} = (1 - 0)/h$ must be an integer, a reasonable choice for the step size is $h = 0.001$. ■

Data Approximation and Neville's Method

- ❖ Previously, we found an explicit representation for Lagrange polynomials and their error when approximating a function on an interval.
- ❖ A frequent use of these polynomials involves the interpolation of tabulated data.
 - ✓ In this case an explicit representation of the polynomial might not be needed, only the values of the polynomial at specified points.
 - ✓ In this situation the function underlying the data might not be known so the explicit form of the error cannot be used.

Data Approximation and Neville's Method

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

- ❖ Suppose we want to approximate the value of the function at $x = 1.5$.
- ❖ The most appropriate linear polynomial uses $x_1 = 1.3$ and $x_2 = 1.6$ because 1.5 is between 1.3 and 1.6.
- ❖ The value of the interpolating polynomial at 1.5 is

$$\begin{aligned}P_1(1.5) &= \frac{(1.5 - 1.6)}{(1.3 - 1.6)} f(1.3) + \frac{(1.5 - 1.3)}{(1.6 - 1.3)} f(1.6) \\&= \frac{(1.5 - 1.6)}{(1.3 - 1.6)} (0.6200860) + \frac{(1.5 - 1.3)}{(1.6 - 1.3)} (0.4554022) \\&= 0.5102968\end{aligned}$$

Data Approximation and Neville's Method

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

❖ Two polynomials of degree 2 can reasonably be used, one with $x_0 = 1.3, x_1 = 1.6$, and $x_2 = 1.9$, which gives

❖ The value of the interpolating polynomial at 1.5 is

$$P_2(1.5) = \frac{(1.5 - 1.6)(1.5 - 1.9)}{(1.3 - 1.6)(1.3 - 1.9)}(0.6200860) + \frac{(1.5 - 1.3)(1.5 - 1.9)}{(1.6 - 1.3)(1.6 - 1.9)}(0.4554022) + \frac{(1.5 - 1.3)(1.5 - 1.6)}{(1.9 - 1.3)(1.9 - 1.6)}(0.2818186) = 0.5112857,$$

❖ $P_2(1.5) = 0.5112857$

❖ and one with $x_0 = 1.0, x_1 = 1.3$, and $x_2 = 1.6$, which gives $\hat{P}_2(1.5) = 0.5124715$.

Data Approximation and Neville's Method

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

- ❖ In the third-degree case, there are also two reasonable choices for the polynomial. One with $x_0 = 1.3$, $x_1 = 1.6$, $x_2 = 1.9$, and $x_3 = 2.2$, which gives $P_3(1.5) = 0.5118302$.
- ❖ The second third-degree approximation is obtained with $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, and $x_3 = 1.9$, which gives $\hat{P}_3(1.5) = 0.5118127$.

Data Approximation and Neville's Method

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

- ❖ The fourth-degree Lagrange polynomial uses all the entries in the table. With $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$, the approximation is $P_4(1.5) = 0.5118200$.

Data Approximation and Neville's Method

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

❖ In summary, we have

$$P_1(1.5) = 0.5102968$$

$$P_2(1.5) = 0.5112857$$

$$\hat{P}_2(1.5) = 0.5124715$$

$$P_3(1.5) = 0.5118302$$

$$\hat{P}_3(1.5) = 0.5118127$$

$$P_4(1.5) = 0.5118200$$

- ❖ Because $P_3(1.5)$, $\hat{P}_3(1.5)$, and $P_4(1.5)$ all agree to within 2×10^{-5} units (safe bound for the differences in values), we expect this degree of accuracy for these approximations.
- ❖ We also expect $P_4(1.5)$ to be the most accurate approximation, since it uses more of the given data.

Data Approximation and Neville's Method

- ❖ The function we are approximating is actually the Bessel function of the first kind of order zero, whose value at 1.5 is known to be 0.5118277.

$$P_1(1.5) = 0.5102968$$

$$P_2(1.5) = 0.5112857$$

$$\hat{P}_2(1.5) = 0.5124715$$

$$P_3(1.5) = 0.5118302$$

$$\hat{P}_3(1.5) = 0.5118127$$

$$P_4(1.5) = 0.5118200$$

$$f(1.5) = 0.5118277$$

$$|P_1(1.5) - f(1.5)| \approx 1.53 \times 10^{-3},$$

$$|P_2(1.5) - f(1.5)| \approx 5.42 \times 10^{-4},$$

$$|\hat{P}_2(1.5) - f(1.5)| \approx 6.44 \times 10^{-4},$$

$$|P_3(1.5) - f(1.5)| \approx 2.5 \times 10^{-6},$$

$$|\hat{P}_3(1.5) - f(1.5)| \approx 1.50 \times 10^{-5},$$

$$|P_4(1.5) - f(1.5)| \approx 7.7 \times 10^{-6}.$$

- ❖ If $f(1.5)$ is known, we accept $P_3(1.5)$. If $f(1.5)$ is unknown, we accept $P_4(1.5)$.

Data Approximation and Neville's Method

- ❖ The function we are approximating is actually the Bessel function of the first kind of order zero, whose value at 1.5 is known to be 0.5118277.

$$P_1(1.5) = 0.5102968$$

$$P_2(1.5) = 0.5112857$$

$$\hat{P}_2(1.5) = 0.5124715$$

$$P_3(1.5) = 0.5118302$$

$$\hat{P}_3(1.5) = 0.5118127$$

$$P_4(1.5) = 0.5118200$$

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$$|P_1(1.5) - f(1.5)| \approx 1.53 \times 10^{-3},$$

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$$|\hat{P}_3(1.5) - f(1.5)| \approx 1.50 \times 10^{-5},$$

$$|P_4(1.5) - f(1.5)| \approx 7.7 \times 10^{-6}.$$

- ❖ The Lagrange error term cannot be applied here ($f^{(4)}$ is unknown). Unfortunately, this is generally the case.

Neville's Method

$$f(x) = P(x) + \frac{f^{(n+1)}(\xi(x))}{(n+1)!} (x - x_0)(x - x_1) \cdots (x - x_n)$$

- ❖ A practical difficulty with Lagrange interpolation is that the error term is difficult to apply.
 - ✓ Suppose you want an error of $\leq 10^{-6}$. Then we would need an n .
 - ✓ But because the error formula contains an **unknown** ξ a **complicated product term**, it is difficult to determine **in advance** what n should be.
- ❖ so the degree of the polynomial needed for the desired accuracy is generally not known until computations have been performed.

Neville's Method

- ❖ A common practice is to compute the results given from various polynomials until appropriate agreement is obtained [as previously illustrated].
- ❖ Calculating the approximation by the second polynomial does not lessen the work needed to calculate the third approximation; nor is the fourth approximation easier to obtain once the third approximation is known, and so on.
- ❖ We will present a formula in a manner that uses the previous calculations to greater advantage.

Neville's Method

Definition: Let f be a function defined at $x_0, x_1, x_2, \dots, x_n$, and suppose that $m_1, m_2, \dots, m_k$ are k distinct integers, with $0 \leq m_i \leq n$ for each i . The Lagrange polynomial that agrees with $f(x)$ at the k points $x_{m_1}, x_{m_2}, \dots, x_{m_k}$ is denoted $P_{m_1, m_2, \dots, m_k}(x)$.

$$P_n(x)$$

✓ *n th Lagrange polynomial*

$$P_{m_1, m_2, \dots, m_k}(x)$$

✓ Lagrange polynomial that agrees with $f(x)$ at the k points
 $x_{m_1}, x_{m_2}, \dots, x_{m_k}$

Neville's Method

Example: Suppose that $x_0 = 1$, $x_1 = 2$, $x_2 = 3$, $x_3 = 4$, $x_4 = 6$, and $f(x) = e^x$. Determine the interpolating polynomial denoted $P_{1,2,4}(x)$, and use this polynomial to approximate $f(5)$.

Solution: This is the Lagrange polynomial that agrees with $f(x)$ at $x_1 = 2$, $x_2 = 3$, and $x_4 = 6$. Hence

$$\begin{aligned}
 P_{1,2,4}(x) &= \frac{(x - 3)(x - 6)}{(2 - 3)(2 - 6)} e^2 + \frac{(x - 2)(x - 6)}{(3 - 2)(3 - 6)} e^3 + \frac{(x - 2)(x - 3)}{(6 - 2)(6 - 3)} e^6 \\
 f(5) \approx P_{1,2,4}(x) &= \frac{(5 - 3)(5 - 6)}{(2 - 3)(2 - 6)} e^2 + \frac{(5 - 2)(5 - 6)}{(3 - 2)(3 - 6)} e^3 + \frac{(5 - 2)(5 - 3)}{(6 - 2)(6 - 3)} e^6 \\
 &= \frac{1}{4} e^2 - \frac{1}{2} e^3 + \frac{1}{2} e^6 \\
 &= \left(\frac{1}{4} e^2 - \frac{1}{2} e^3 + \frac{1}{2} e^6 \right) x^2 + \left(\frac{1}{2} e^2 - 4 e^3 + \frac{1}{2} e^6 \right) x + \left(\frac{1}{4} e^2 - \frac{1}{2} e^3 + \frac{1}{2} e^6 \right)
 \end{aligned}$$

Neville's Method

Example: Suppose that $x_0 = 1$, $x_1 = 2$, $x_2 = 3$, $x_3 = 4$, $x_4 = 6$, and $f(x) = e^x$. Determine the interpolating polynomial denoted $P_{1,2,4}(x)$, and use this polynomial to approximate $f(5)$.

Solution: This is the Lagrange polynomial that agrees with $f(x)$ at $x_1 = 2$, $x_2 = 3$, and $x_4 = 6$. Hence

$$P_{1,2,4}(x) = \frac{(x-3)(x-6)}{(2-3)(2-6)}e^2 + \frac{(x-2)(x-6)}{(3-2)(3-6)}e^3 + \frac{(x-2)(x-3)}{(6-2)(6-3)}e^6$$

$$\begin{aligned} f(5) \approx P_{1,2,4}(x) &= \frac{(5-3)(5-6)}{(2-3)(2-6)}e^2 + \frac{(5-2)(5-6)}{(3-2)(3-6)}e^3 + \frac{(5-2)(5-3)}{(6-2)(6-3)}e^6 \\ &= -\frac{1}{2}e^2 + e^3 + \frac{1}{2}e^6 \approx 218.105. \end{aligned}$$

Neville's Method

Example: Suppose that $x_0 = 1$, $x_1 = 2$, $x_2 = 3$, $x_3 = 4$, $x_4 = 6$, and $f(x) = e^x$. Determine the interpolating polynomial denoted $P_{1,2,4}(x)$, and use this polynomial to approximate $f(5)$.

Solution: This is the Lagrange polynomial that agrees with $f(x)$ at $x_1 = 2$, $x_2 = 3$, and $x_4 = 6$. Hence

$$P_{1,2,4}(x) = \frac{(x-3)(x-6)}{(2-3)(2-6)}e^2 + \frac{(x-2)(x-6)}{(3-2)(3-6)}e^3 + \frac{(x-2)(x-3)}{(6-2)(6-3)}e^6$$

$$\begin{aligned} f(5) \approx P_{1,2,4}(x) &= \frac{(5-3)(5-6)}{(2-3)(2-6)}e^2 + \frac{(5-2)(5-6)}{(3-2)(3-6)}e^3 + \frac{(5-2)(5-3)}{(6-2)(6-3)}e^6 \\ &= -\frac{1}{2}e^2 + e^3 + \frac{1}{2}e^6 \approx 218.105. \end{aligned}$$

Neville's Method

Theorem: Let f be defined at $x_0, x_1, \dots, x_k$ and let x_j and x_i be two distinct numbers in this set. Then

$$P(x) = \frac{(x - x_j)P_{0,1,2,\dots,j-1,j+1,\dots,k}(x) - (x - x_i)P_{0,1,2,\dots,i-1,i+1,\dots,k}(x)}{x_i - x_j}$$

is the k th Lagrange polynomial that interpolates f at $k + 1$ points $x_0, x_1, \dots, x_k$.

✓ The theorem implies that the interpolating polynomials can be generated recursively.

✓ Example:

$$P_{0,1} = \frac{(x-x_0)P_1 - (x-x_1)P_0}{(x_1-x_0)}, P_{1,2} = \frac{(x-x_1)P_2 - (x-x_2)P_1}{(x_2-x_1)}, P_{0,1,2} = \frac{(x-x_0)P_{1,2} - (x-x_2)P_{0,1}}{(x_2-x_0)}$$

Neville's Method

- ❖ The procedure that uses the result of the previous theorem to recursively generate interpolating polynomial approximations is called Neville's method.

x_0	P_0				
x_1	P_1	$P_{0,1}$			
x_2	P_2	$P_{1,2}$	$P_{0,1,2}$		
x_3	P_3	$P_{2,3}$	$P_{1,2,3}$	$P_{0,1,2,3}$	
x_4	P_4	$P_{3,4}$	$P_{2,3,4}$	$P_{1,2,3,4}$	$P_{0,1,2,3,4}$

- ✓ The P notation is cumbersome because of the number of subscripts used to represent the entries. Note, however, that as an array is being constructed, only two subscripts are needed.

Neville's Method

- ✓ The P notation is cumbersome because of the number of subscripts used to represent the entries. Note, however, that as an array is being constructed, only two subscripts are needed.
- ✓ To avoid the multiple subscripts, we let $Q_{i,j}(x)$, for $0 \leq j \leq i$, denote the interpolating polynomial of degree j on the $(j + 1)$ numbers $x_{i-j}, x_{i-j+1}, \dots, x_{i-1}, x_i$; that is,

x_0	$P_0 = Q_{0,0}$					
x_1	$P_1 = Q_{1,0}$	$P_{0,1} = Q_{1,1}$				
x_2	$P_2 = Q_{2,0}$	$P_{1,2} = Q_{2,1}$	$P_{0,1,2} = Q_{2,2}$			
x_3	$P_3 = Q_{3,0}$	$P_{2,3} = Q_{3,1}$	$P_{1,2,3} = Q_{3,2}$	$P_{0,1,2,3} = Q_{3,3}$		
x_4	$P_4 = Q_{4,0}$	$P_{3,4} = Q_{4,1}$	$P_{2,3,4} = Q_{4,2}$	$P_{1,2,3,4} = Q_{4,3}$	$P_{0,1,2,3,4} = Q_{4,4}$	

Neville's Method

Example: Apply Neville's method to the data (on the right) by constructing a recursive table to interpolate the value of the function at $x = 1.5$

Solution: Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$.

$$Q_{0,0} = f(1.0) = 0.7651977$$

$$Q_{1,0} = f(1.3) = 0.6200860$$

$$Q_{2,0} = f(1.6) = 0.4554022$$

$$Q_{3,0} = f(1.9) = 0.2818186$$

$$Q_{4,0} = f(2.2) = 0.1103623$$

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

Neville's Method

Example: Apply Neville's method to the data (on the right) by constructing a recursive table to interpolate the value of the function at $x = 1.5$

Solution: Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$.

$$Q_{1,1} = \frac{(x - x_0)Q_{1,0} - (x - x_1)Q_{0,0}}{x_1 - x_0}$$

$$\begin{aligned} Q_{1,1}(1.5) &= \frac{(1.5 - 1.0)(0.6200860) - (1.5 - 1.3)(0.7651977)}{1.3 - 1.0} \\ &= 0.5233449 \end{aligned}$$

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

Neville's Method

Example: Apply Neville's method to the data (on the right) by constructing a recursive table to interpolate the value of the function at $x = 1.5$

Solution: Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$.

$$Q_{1,1}(1.5) = 0.5233449$$

$$Q_{2,1}(1.5) = 0.5102968$$

$$Q_{3,1}(1.5) = 0.5132634$$

$$Q_{4,1}(1.5) = 0.5104270$$

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

✓ The best linear approximation is expected to be $Q_{2,1}$ because 1.5 is between $x_1 = 1.3$ and $x_2 = 1.6$.

Neville's Method

Example: Apply Neville's method to the data (on the right) by constructing a recursive table to interpolate the value of the function at $x = 1.5$

Solution: Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$.

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

$$Q_{2,2} = \frac{(x - x_0)Q_{2,1} - (x - x_2)Q_{1,1}}{x_2 - x_0}$$

x_1, x_2 missing x_0

x_0, x_1 missing x_2

$$\begin{aligned} Q_{2,2}(1.5) &= \frac{(1.5 - 1.0)(0.5102968) - (1.5 - 1.6)(0.5233449)}{1.6 - 1.0} \\ &= 0.5124715 \end{aligned}$$

Neville's Method

Example: Apply Neville's method to the data (on the right) by constructing a recursive table to interpolate the value of the function at $x = 1.5$

Solution: Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$.

$$Q_{2,2}(1.5) = 0.5124715$$

$$Q_{3,2}(1.5) = 0.5112857$$

$$Q_{4,2}(1.5) = 0.5137361$$

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

Neville's Method

Example: Apply Neville's method to the data (on the right) by constructing a recursive table to interpolate the value of the function at $x = 1.5$

Solution: Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$.

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

	deg 0	deg 1	deg 2	deg 3	deg 4
1.0	0.7651977				
1.3	0.6200860	0.5233449			
1.6	0.4554022	0.5102968	0.5124715		
1.9	0.2818186	0.5132634	0.5112857	0.5118127	
2.2	0.1103623	0.5104270	0.5137361	0.5118302	0.5118200

Neville's Method

Example: Apply Neville's method to the data (on the right) by constructing a recursive table to interpolate the value of the function at $x = 1.5$

If the latest approximation, $Q_{4,4}$, was not sufficiently accurate, another node, x_5 , could be selected, and another row added to the table:

$$x_5 \quad Q_{5,0} \quad Q_{5,1} \quad Q_{5,2} \quad Q_{5,3} \quad Q_{5,4} \quad Q_{5,5}.$$

Then $Q_{4,4}$, $Q_{5,4}$, and $Q_{5,5}$ could be compared to determine further accuracy.

The function in Example 2 is the Bessel function of the first kind of order zero, whose value at 2.5 is -0.0483838 , and the next row of approximations to $f(1.5)$ is

$$2.5 \quad -0.0483838 \quad 0.4807699 \quad 0.5301984 \quad 0.5119070 \quad 0.5118430 \quad 0.5118277.$$

The final new entry, 0.5118277 , is correct to all seven decimal places.

Neville's Method

Example: The table on the right lists the values of $f(x) = \ln x$ accurate to the places given. Use Neville's method and four-digit rounding arithmetic to approximate $f(2.1) = \ln 2.1$ by completing the Neville.

Solution: Let $x_0 = 2.0, x_1 = 2.2, x_2 = 2.3$.

i	x_i	$\ln x_i$
0	2.0	0.6931
1	2.2	0.7885
2	2.3	0.8329

i	x_i	$x - x_i$	$Q_{i,0}$	$Q_{i,1}$	$Q_{i,2}$
0	2.0	0.1	0.6931		
1	2.2	-0.1	0.7885	0.7410	
2	2.3	-0.2	0.8329	0.7441	0.7420

In the preceding example we have $f(2.1) = \ln 2.1 = 0.7419$ to four decimal places, so the absolute error is

$$|f(2.1) - P_2(2.1)| = |0.7419 - 0.7420| = 10^{-4}.$$

However, $f'(x) = 1/x$, $f''(x) = -1/x^2$, and $f'''(x) = 2/x^3$, so the Lagrange error formula (3.3) in Theorem 3.3 gives the error bound

$$\begin{aligned} |f(2.1) - P_2(2.1)| &= \left| \frac{f'''(\xi(2.1))}{3!} (x - x_0)(x - x_1)(x - x_2) \right| \\ &= \left| \frac{1}{3(\xi(2.1))^3} (0.1)(-0.1)(-0.2) \right| \leq \frac{0.002}{3(2)^3} = 8.\bar{3} \times 10^{-5}. \end{aligned}$$

Notice that the actual error, 10^{-4} , exceeds the error bound, $8.\bar{3} \times 10^{-5}$. This apparent contradiction is a consequence of finite-digit computations. We used four-digit rounding arithmetic, and the Lagrange error formula (3.3) assumes infinite-digit arithmetic. This caused our actual errors to exceed the theoretical error estimate.

Newton's Divided Difference

- ❖ Suppose that $P_n(x)$ is the n th Lagrange polynomial that agrees with the function f at the distinct numbers $x_0, x_1, \dots, x_n$.
- ❖ Although this Lagrange polynomial is unique, there are alternate algebraic representations that are useful in certain situations.
- ❖ The divided differences of f with respect to $x_0, x_1, \dots, x_n$ are used to express $P_n(x)$ in the form

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \dots + a_n(x - x_0) \cdots (x - x_{n-1})$$

$$P_4(x) = \underbrace{(a_0)}_{\text{for appropriate constants } a_0, a_1, \dots, a_n} + \underbrace{(a_1)(x - x_0)}_{\text{for appropriate constants } a_0, a_1, \dots, a_n} + \underbrace{(a_2)(x - x_0)(x - x_1)}_{\text{for appropriate constants } a_0, a_1, \dots, a_n} + \underbrace{(a_3)(x - x_0)(x - x_1)(x - x_2)}_{\text{for appropriate constants } a_0, a_1, \dots, a_n} + \underbrace{(a_4)(x - x_0)(x - x_1)(x - x_2)(x - x_3)}_{\text{for appropriate constants } a_0, a_1, \dots, a_n}$$

- ❖ Our objective is to identify the constants $a_0, a_1, \dots, a_n$.

Newton's Divided Difference

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0) \cdots (x - x_{n-1})$$

- ✓ Observe that when we evaluate the function $P_n(x)$ at $x = x_0$, we'll have

$$P_n(x_0) = a_0 = f(x_0)$$

- ✓ Also, when we evaluate the function $P_n(x)$ at $x = x_1$, we'll have

$$P_n(x_1) = a_0 + a_1(x_1 - x_0) = f(x_1)$$

$$f(x_0) + a_1(x_1 - x_0) = f(x_1)$$

$$a_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

Newton's Divided Difference

$$P_n(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + \cdots + a_n(x - x_0) \cdots (x - x_{n-1})$$

✓ Observe that when we evaluate the function $P_n(x)$ at $x = x_2$, we'll have

$$P_n(x_2) = a_0 + a_1(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1) = f(x_2)$$

$$f(x_2) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_0) + a_2(x_2 - x_0)(x_2 - x_1)$$

$$\frac{f(x_2) - f(x_0) - \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)} = a_2$$

Newton's Divided Difference

$$\begin{aligned}a_2 &= \frac{f(x_2) - f(x_0) - \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)} \\&= \frac{f(x_2) - f(x_1) + (f(x_1) - f(x_0)) - \frac{f(x_1) - f(x_0)}{x_1 - x_0} (x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)} \\&= \frac{f(x_2) - f(x_1) + (f(x_1) - f(x_0)) \left(1 - \frac{x_2 - x_0}{x_1 - x_0}\right)}{(x_2 - x_0)(x_2 - x_1)} \\&= \frac{f(x_2) - f(x_1) + (f(x_1) - f(x_0)) \left(\frac{x_1 - x_2}{x_1 - x_0}\right)}{(x_2 - x_0)(x_2 - x_1)}\end{aligned}$$

Newton's Divided Difference

$$\begin{aligned} &= \frac{f(x_2) - f(x_1) + (f(x_1) - f(x_0)) \left(\frac{x_1 - x_2}{x_1 - x_0} \right)}{(x_2 - x_0)(x_2 - x_1)} \\ &= \frac{f(x_2) - f(x_1)}{(x_2 - x_0)(x_2 - x_1)} + \frac{(f(x_1) - f(x_0)) \left(\frac{x_1 - x_2}{x_1 - x_0} \right)}{(x_2 - x_0)(x_2 - x_1)} \\ &= \frac{f(x_2) - f(x_1)}{(x_2 - x_0)(x_2 - x_1)} - \frac{(f(x_1) - f(x_0)) \left(\frac{1}{x_1 - x_0} \right)}{(x_2 - x_0)} \\ a_2 &= \frac{1}{x_2 - x_0} \left[\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] \end{aligned}$$

Newton's Divided Difference

❖ We now introduce the following notation:

✓ zeroth divided difference

$$f[x_i] = f(x_i)$$

✓ first divided difference

$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$$

✓ second divided difference

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

✓ k th divided difference relative to $x_i, x_{i+1}, x_{i+2}, \dots, x_{i+k}$ is

$$f[x_i, x_{i+1}, x_{i+2}, \dots, x_{i+k}] = \frac{f[x_{i+1}, x_{i+2}, \dots, x_{i+k}] - f[x_i, x_{i+1}, \dots, x_{i+k-1}]}{x_{i+k} - x_i}$$

Newton's Divided Difference

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✓ zeroth divided difference

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$$f[x_i, x_{i+1}] = \frac{f[x_{i+1}] - f[x_i]}{x_{i+1} - x_i}$$

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$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

✓ k th divided difference relative to $x_0, x_1, x_2, \dots, x_k$ is

$$f[x_0, x_1, x_2, \dots, x_k] = \frac{f[x_1, x_2, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}$$

Newton's Divided Difference

❖ Based on our previous results, we have

$$f[x_0] = f(x_0)$$

$$a_0 = f(x_0)$$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$$

$$a_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

$$a_2 = \frac{\left[\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0} \right]}{x_2 - x_0}$$

$$f[x_0, x_1, x_2, \dots, x_k] = \frac{f[x_1, x_2, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}$$

Newton's Divided Difference

❖ We now have the Newton's divided difference formula given by

$$P_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, x_1, x_2, \dots, x_k](x - x_0)(x - x_1) \dots (x - x_{k-1})$$

Newton's Divided Difference

x	$f(x)$	First divided differences	Second divided differences	Third divided differences
x_0	$f[x_0]$	$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0}$		
x_1	$f[x_1]$	$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1}$	$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$	$f[x_0, x_1, x_2, x_3] = \frac{f[x_1, x_2, x_3] - f[x_0, x_1, x_2]}{x_3 - x_0}$
x_2	$f[x_2]$	$f[x_2, x_3] = \frac{f[x_3] - f[x_2]}{x_3 - x_2}$	$f[x_1, x_2, x_3] = \frac{f[x_2, x_3] - f[x_1, x_2]}{x_3 - x_1}$	$f[x_1, x_2, x_3, x_4] = \frac{f[x_2, x_3, x_4] - f[x_1, x_2, x_3]}{x_4 - x_1}$
x_3	$f[x_3]$	$f[x_3, x_4] = \frac{f[x_4] - f[x_3]}{x_4 - x_3}$	$f[x_2, x_3, x_4] = \frac{f[x_3, x_4] - f[x_2, x_3]}{x_4 - x_2}$	$f[x_2, x_3, x_4, x_5] = \frac{f[x_3, x_4, x_5] - f[x_2, x_3, x_4]}{x_5 - x_2}$
x_4	$f[x_4]$	$f[x_4, x_5] = \frac{f[x_5] - f[x_4]}{x_5 - x_4}$	$f[x_3, x_4, x_5] = \frac{f[x_4, x_5] - f[x_3, x_4]}{x_5 - x_3}$	
x_5	$f[x_5]$			

Newton's Divided Difference

Example: Complete the divided difference table for the data on the right, and construct the interpolating polynomial that uses all this data.

Solution: Let $x_0 = 1.0, x_1 = 1.3, x_2 = 1.6, x_3 = 1.9$ and $x_4 = 2.2$.

x	$f(x)$
1.0	0.7651977
1.3	0.6200860
1.6	0.4554022
1.9	0.2818186
2.2	0.1103623

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
1	1.3	0.6200860	-0.4837057			
2	1.6	0.4554022	-0.5489460	-0.1087339	0.0658784	
3	1.9	0.2818186	-0.5786120	-0.0494433	0.0680685	0.0018251
4	2.2	0.1103623	-0.5715210	0.0118183		

$$f[x_0, x_1] = \frac{0.6200860 - 0.7651977}{1.3 - 1.0}$$

Newton's Divided Difference

Example: Complete the divided difference table for the data on the right, and construct the interpolating polynomial that uses all this data.

Solution: Let $x_0 = 1.0$, $x_1 = 1.3$, $x_2 = 1.6$, $x_3 = 1.9$ and $x_4 = 2.2$.

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
1	1.3	0.6200860	-0.4837057			
2	1.6	0.4554022	-0.5489460	-0.1087339	0.0658784	
3	1.9	0.2818186	-0.5786120	-0.0494433	0.0680685	0.0018251
4	2.2	0.1103623	-0.5715210	0.0118183		

Answer: $P(x) = 0.7651977 - 0.4837057(x - 1.0) - 0.1087339(x - 1.0)(x - 1.3) + 0.0658784(x - 1.0)(x - 1.3)(x - 1.6) + 0.0018251(x - 1.0)(x - 1.3)(x - 1.6)(x - 1.9).$

Newton Forward-Difference Formula

Definition: For a given sequence $\{p_n\}_{n=0}^{\infty}$, the forward difference Δp_n (read “delta p_n ”) is defined by

$$\Delta p_n = p_{n+1} - p_n, \text{ for } n \geq 0$$

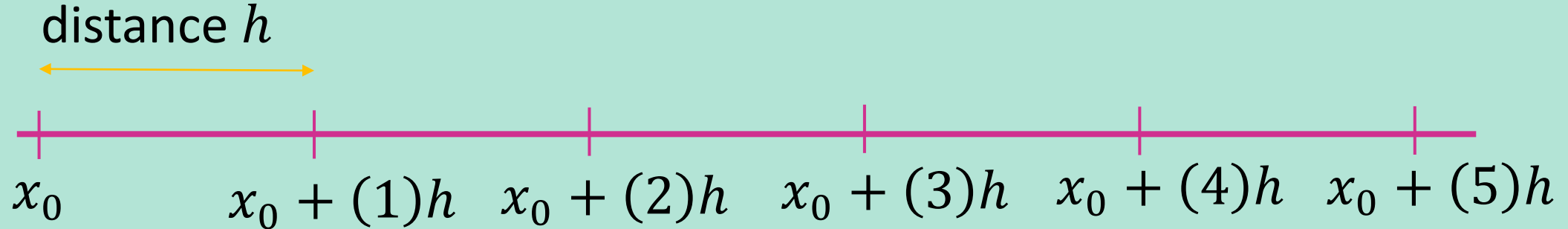
Higher powers of the operator Δ are defined recursively by $\Delta^k p_n = \Delta(\Delta^{k-1} p_n)$, for $k \geq 2$

✓ The definition implies that

$$\begin{aligned}\Delta^2 p_n &= \Delta(p_{n+1} - p_n) \\ &= \Delta p_{n+1} - \Delta p_n \\ &= (p_{n+2} - p_{n+1}) - (p_{n+1} - p_n) \\ &= p_{n+2} - 2p_{n+1} + p_n.\end{aligned}$$

Newton Forward-Difference Formula

- ❖ Newton's divided-difference formula can be expressed in a simplified form when the nodes are arranged consecutively with equal spacing.
- ✓ We introduce the notation $h = x_{i+1} - x_i$, for each $i = 0, 1, \dots, n - 1$ and let $x = x_0 + sh$ wherein s is a nonnegative integer.



Newton Forward-Difference Formula

- ❖ Newton's divided-difference formula can be expressed in a simplified form when the nodes are arranged consecutively with equal spacing.
- ✓ We introduce the notation $h = x_{i+1} - x_i$, for each $i = 0, 1, \dots, n - 1$ and let $x = x_0 + sh$ wherein s is a nonnegative integer.
- ✓ It follows that

$$x = x_0 + sh$$

$$x - x_1 = x_0 + sh - x_1$$

$$x - x_1 = x_0 + sh - (h + x_0)$$

$$x - x_1 = x_0 + sh - h - x_0$$

$$x - x_1 = (s - 1)h$$

Newton Forward-Difference Formula

- ❖ Newton's divided-difference formula can be expressed in a simplified form when the nodes are arranged consecutively with equal spacing.
- ✓ We introduce the notation $h = x_{i+1} - x_i$, for each $i = 0, 1, \dots, n - 1$ and let $x = x_0 + sh$ wherein s is a nonnegative integer.
- ✓ It follows that

$$x = x_0 + sh$$

$$x - x_2 = x_0 + sh - x_2$$

$$x - x_2 = x_0 + sh - (h + x_1)$$

$$x - x_2 = x_0 + sh - h - (h + x_0)$$

$$x - x_2 = (s - 2)h$$

Newton Forward-Difference Formula

- ❖ Newton's divided difference formula can be expressed in a simplified form when the nodes are arranged consecutively with equal spacing.
- ✓ We introduce the notation $h = x_{i+1} - x_i$, for each $i = 0, 1, \dots, n - 1$ and let $x = x_0 + sh$ wherein s is not zero.
- ✓ It follows that

$$x - x_i = (s - i)h$$

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) \\ + \dots + f[x_0, x_1, \dots, x_n](x - x_0) \dots (x - x_{n-1})$$

$$P_n(x_0 + sh) = f[x_0] + f[x_0, x_1](sh) + f[x_0, x_1, x_2](sh)((s - 1)h) \\ + \dots + f[x_0, x_1, \dots, x_n](sh)((s - 1)h) \dots ((s - (n - 1))h)$$

Newton Forward-Difference Formula

$$P_n(x) = f[x_0] + f[x_0, x_1](sh) + f[x_0, x_1, x_2](sh)((s-1)h) \\ + \cdots + f[x_0, x_1, \dots, x_{n-1}](sh)((s-1)h) \cdots ((s-(n-1))h)$$

$$P_n(x) = f[x_0] + \sum_{k=1}^n f[x_1, x_2, \dots, x_k](sh)((s-1)h) \cdots ((s-(k-1))h)$$

$$P_n(x) = f[x_0] + \sum_{k=1}^n f[x_1, x_2, \dots, x_k](s)((s-1)) \cdots ((s-(k-1))) h^k$$

$$P_n(x) = f[x_0] + \sum_{k=1}^n \binom{s}{k} k! h^k f[x_1, x_2, \dots, x_k]$$

$$\text{wherein } \binom{s}{k} = \frac{s(s-1)(s-2) \cdots (s-k+1)}{k!}$$

Newton Forward-Difference Formula

❖ We now have

$$P_n(x) = P_n(x_0 + sh) = f[x_0] + \sum_{k=1}^n \binom{s}{k} k! h^k f[x_0, x_1, \dots, x_k]$$

wherein $\binom{s}{k} = \frac{s(s-1)(s-2) \dots (s-k+1)}{k!}$

Newton Forward-Difference Formula

❖ Using the forward difference notation $\Delta p_n = p_{n+1} - p_n$, we will get

$$f[x_0, x_1] = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{1}{h} (f(x_1) - f(x_0)) = \frac{1}{h} \Delta f(x_0)$$

$$\begin{aligned} f[x_0, x_1, x_2] &= \frac{\left[\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0} \right]}{x_2 - x_0} = \frac{1}{2h} \left(\frac{\Delta f(x_1)}{h} - \frac{\Delta f(x_0)}{h} \right) \\ &= \frac{1}{2h^2} \Delta^2 f(x_0) \end{aligned}$$

❖ In general, we have

$$f[x_0, x_1, x_2, \dots, x_k] = \frac{1}{k! h^k} \Delta^k f(x_0)$$

Newton Forward-Difference Formula

❖ Since we have

$$f[x_0, x_1, x_2, \dots, x_k] = \frac{1}{k! h^k} \Delta^k f(x_0)$$

then we can rewrite

$$P_n(x) = P_n(x_0 + sh) = f[x_0] + \sum_{k=1}^n \binom{s}{k} k! h^k f[x_0, x_1, \dots, x_k]$$

into the **Newton forward-difference formula**

$$P_n(x) = P_n(x_0 + sh) = f[x_0] + \sum_{k=1}^n \binom{s}{k} \Delta^k f(x_0)$$

Newton Forward-Difference Formula

Example: Consider the following divided difference table we previously obtained.

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
1	1.3	0.6200860	-0.4837057			
2	1.6	0.4554022	-0.5489460	-0.1087339	0.0658784	
3	1.9	0.2818186	-0.5786120	-0.0494433	0.0680685	0.0018251
4	2.2	0.1103623	-0.5715210	0.0118183		

We will interpolate f at $x = 1.1$ and $x = 2$ using the given five points.

Solution:

Given:

$$x = 1.1 = 1 + 0.1$$

$$h = 1.3 - 1.0 = 0.3$$

so $s = \frac{1}{3}$, that is

$$x = 1.1 = 1 + \frac{1}{3}(0.3)$$

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977	-0.4837057			
1	1.3	0.6200860	-0.5489460	-0.1087339	0.0658784	
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3	1.9	0.2818186	-0.5715210	0.0118183		
4	2.2	0.1103623				

Using Newton's Divided difference formula:

$$\begin{aligned} \checkmark P_n(x_0 + sh) = & f[x_0] + f[x_0, x_1](sh) + f[x_0, x_1, x_2](sh)((s-1)h) \\ & + \dots + f[x_0, x_1, \dots, x_n](sh)((s-1)h) \dots ((s-(n-1))h) \end{aligned}$$

Using Newton's Divided difference formula:

$$P_4\left(1 + \frac{1}{3}(0.3)\right) = 0.7196459912$$

Using Newton's Divided difference formula:

$$P_4 \left(1 + \frac{1}{3}(0.3) \right)$$

$$\begin{aligned}
 &= \overset{f[x_0]}{0.7651977} + \overset{f[x_0, x_1]}{(-0.4837057)} \left(\frac{1}{3} \right) \overset{h}{(0.3)} \\
 &\quad + \overset{f[x_0, x_1, x_2]}{(-0.1087339)} \left(\frac{1}{3} \right) \overset{h}{(0.3)} \left(\left(\frac{1}{3} - 1 \right) \overset{h}{(0.3)} \right) \\
 &\quad + \overset{f[x_0, x_1, x_2, x_3]}{(0.0658784)} \left(\frac{1}{3} \right) \overset{h}{(0.3)} \left(\left(\frac{1}{3} - 1 \right) \overset{h}{(0.3)} \right) \left(\left(\frac{1}{3} - 2 \right) \overset{h}{(0.3)} \right) \\
 &\quad + \overset{f[x_0, x_1, x_2, x_3, x_4]}{(0.0018251)} \left(\frac{1}{3} \right) \overset{h}{(0.3)} \left(\left(\frac{1}{3} - 1 \right) \overset{h}{(0.3)} \right) \left(\left(\frac{1}{3} - 2 \right) \overset{h}{(0.3)} \right) \left(\left(\frac{1}{3} - 3 \right) \overset{h}{(0.3)} \right)
 \end{aligned}$$

Handwritten notes: h is circled in blue in each term. Blue arrows point from h to the 0.3 values. The term 0.3^2 is written in blue next to the third term, and 0.3^3 is written in blue next to the fourth term.

$$= 0.7196459912$$

Solution:

Given:

$$x = 1.1 = 1 + 0.1$$

$$h = 1.3 - 1.0 = 0.3$$

so $s = \frac{1}{3}$, that is

$$x = 1.1 = 1 + \frac{1}{3}(0.3)$$

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
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4	2.2	0.1103623	-0.5715210	0.0118183		

Using Newton's Divided difference formula:

$$\checkmark P_n(x_0 + sh) = f[x_0] + \sum_{k=1}^n \binom{s}{k} k! h^k f[x_0, x_1, \dots, x_k]$$

$s(s-1)(s-2) \dots$

$$P_4\left(1 + \frac{1}{3}(0.3)\right) = f[x_0] + \sum_{k=1}^4 \binom{1/3}{k} k! (0.3)^k f[x_0, x_1, \dots, x_k]$$

$$\begin{aligned}
P_4 \left(1 + \frac{1}{3} (0.3) \right) &= f[x_0] + \sum_{k=1}^4 \binom{1/3}{k} k! (0.3)^k f[x_0, x_1, \dots, x_k] \\
&= f[x_0] + \binom{1/3}{1} 1! (0.3)^1 f[x_0, x_1] + \binom{1/3}{2} 2! (0.3)^2 f[x_0, x_1, x_2] \\
&\quad + \binom{1/3}{3} 3! (0.3)^3 f[x_0, x_1, x_2, x_3] + \binom{1/3}{4} 4! (0.3)^4 f[x_0, x_1, x_2, x_3, x_4] \\
&= 0.7651977 + \binom{1}{\frac{1}{3}} 1! (0.3)^1 (-0.4837057) + \binom{1}{\frac{1}{3}} 2! (0.3)^2 (-0.1087339) \\
&\quad + \binom{1}{\frac{1}{3}} 3! (0.3)^3 (0.0658784) + \binom{1}{\frac{1}{3}} 4! (0.3)^4 (0.0018251)
\end{aligned}$$

$$P_4 \left(1 + \frac{1}{3}(0.3) \right) = f[x_0] + \sum_{k=1}^4 \binom{1/3}{k} k! (0.3)^k f[x_0, x_1, \dots, x_k]$$

$$= 0.7651977 + (-0.4837057) \left(\frac{1}{3} \right) (0.3)$$

$$+ (-0.1087339) \left(\frac{1}{3} \right) \left(\frac{1}{3} - 1 \right) (0.3)^2 + (0.0658784) \left(\frac{1}{3} \right) \left(\frac{1}{3} - 1 \right) \left(\frac{1}{3} - 2 \right) (0.3)^3$$

$$+ (0.0018251) \left(\frac{1}{3} \right) \left(\frac{1}{3} - 1 \right) \left(\frac{1}{3} - 2 \right) \left(\frac{1}{3} - 3 \right) (0.3)^4$$

$$= 0.7196459912$$

Solution:

Given:

$$x = 1.1 = 1 + 0.1$$

$$h = 1.3 - 1.0 = 0.3$$

so $s = \frac{1}{3}$, that is

$$x = 1.1 = 1 + \frac{1}{3}(0.3)$$

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
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2	1.6	0.4554022		-0.0494433		0.0018251
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3	1.9	0.2818186		0.0118183		
			-0.5715210			
4	2.2	0.1103623				

Using Newton forward-difference formula:

$$P_n(x_0 + sh) = f[x_0] + \sum_{k=1}^n \binom{s}{k} \Delta^k f(x_0)$$

$$P_4\left(1 + \frac{1}{3}(0.3)\right) = f[x_0] + \sum_{k=1}^4 \binom{\frac{1}{3}}{k} \Delta^k f(x_0)$$

$$\begin{aligned}
 P_4 \left(1 + \frac{1}{3}(0.3) \right) &= f[x_0] + \sum_{k=1}^4 \binom{1}{\frac{1}{3}} \Delta^k f(x_0) \\
 &= f[x_0] + \binom{1}{\frac{1}{3}} \Delta^1 f(x_0) + \binom{1}{\frac{1}{3}} \Delta^2 f(x_0) + \binom{1}{\frac{1}{3}} \Delta^3 f(x_0) + \binom{1}{\frac{1}{3}} \Delta^4 f(x_0)
 \end{aligned}$$

Note that

- ✓ $\Delta^1 f(x_0) = f(x_1) - f(x_0)$
- ✓ $\Delta^2 f(x_0) = \Delta f(x_1) - \Delta f(x_0) = f(x_2) - 2f(x_1) + f(x_0)$
- ✓ $\Delta^3 f(x_0) = \Delta f(x_2) - \Delta(2f(x_1)) + \Delta f(x_0)$
 $= f(x_3) - f(x_2) - (2f(x_2) - 2f(x_1)) + f(x_1) - f(x_0)$
 $= f(x_3) - 3f(x_2) + 3f(x_1) - f(x_0)$
- ✓ $\Delta^4 f(x_0) = f(x_4) - 4f(x_3) + 6f(x_2) - 4f(x_1) + f(x_0)$

$$P_4 \left(1 + \frac{1}{3}(0.3) \right) = f[x_0] + \sum_{k=1}^4 \binom{1}{k} \Delta^k f(x_0)$$

$$= f[x_0] + \binom{1}{1} \Delta^1 f(x_0) + \binom{1}{2} \Delta^2 f(x_0) + \binom{1}{3} \Delta^3 f(x_0) + \binom{1}{4} \Delta^4 f(x_0)$$

Note that

- ✓ $\Delta^1 f(x_0) = f(x_1) - f(x_0)$
 $= 0.6200860 - 0.7651977 = -0.1451117$
- ✓ $\Delta^2 f(x_0) = f(x_2) - 2f(x_1) + f(x_0)$
 $= 0.4554022 - 2(0.6200860) + 0.7651977 = -0.0195721$
- ✓ $\Delta^3 f(x_0) = f(x_3) - 3f(x_2) + 3f(x_1) - f(x_0)$
 $= 0.2818186 - 3(0.4554022) + 3(0.6200860) - 0.7651977 = 0.0106723$
- ✓ $\Delta^4 f(x_0) = f(x_4) - 4f(x_3) + 6f(x_2) - 4f(x_1) + f(x_0)$
 $= 0.1103623 - 4(0.2818186) + 6(0.4554022) - 4(0.6200860) + 0.7651977$
 $= 0.0003548$

$$\begin{aligned}
 P_4 \left(1 + \frac{1}{3}(0.3) \right) &= f[x_0] + \sum_{k=1}^4 \binom{1}{\frac{1}{3}} \Delta^k f(x_0) \\
 &= f[x_0] + \binom{1}{\frac{1}{3}} \Delta^1 f(x_0) + \binom{1}{\frac{1}{3}} \Delta^2 f(x_0) + \binom{1}{\frac{1}{3}} \Delta^3 f(x_0) + \binom{1}{\frac{1}{3}} \Delta^4 f(x_0)
 \end{aligned}$$

Note that

$$✓ \Delta^1 f(x_0) = -0.1451117$$

$$✓ \Delta^2 f(x_0) = -0.0195721$$

$$✓ \Delta^3 f(x_0) = 0.0106723$$

$$✓ \Delta^4 f(x_0) = 0.0003548$$

$$= 0.7651977 + \binom{1}{\frac{1}{3}} (-0.1451117) + \binom{1}{\frac{1}{3}} (-0.0195721)$$

$$+ \binom{1}{\frac{1}{3}} (0.0106723) + \binom{1}{\frac{1}{3}} (0.0003548)$$

$$P_4 \left(1 + \frac{1}{3} (0.3) \right)$$

$$\binom{s}{k} = \frac{s(s-1)(s-2) \dots (s-k+1)}{k!}$$

$$= 0.7651977 + \binom{\frac{1}{3}}{1} (-0.1451117) + \binom{\frac{1}{3}}{2} (-0.0195721)$$

$$+ \binom{\frac{1}{3}}{3} (0.0106723) + \binom{\frac{1}{3}}{4} (0.0003548)$$

$$= 0.7651977 + \binom{\frac{1}{3}}{1} (-0.1451117) + \frac{\frac{1}{3} \left(\frac{1}{3} - 1 \right)}{2!} (-0.0195721)$$

$$+ \frac{\frac{1}{3} \left(\frac{1}{3} - 1 \right) \left(\frac{1}{3} - 2 \right)}{3!} (0.0106723)$$

$$+ \frac{\frac{1}{3} \left(\frac{1}{3} - 1 \right) \left(\frac{1}{3} - 2 \right) \left(\frac{1}{3} - 3 \right)}{4!} (0.0003548)$$

$$\begin{aligned}
& P_4 \left(1 + \frac{1}{3}(0.3) \right) \\
&= 0.7651977 + \left(\frac{1}{3} \right) (-0.1451117) + \frac{\frac{1}{3} \left(\frac{1}{3} - 1 \right)}{2!} (-0.0195721) \\
&+ \frac{\frac{1}{3} \left(\frac{1}{3} - 1 \right) \left(\frac{1}{3} - 2 \right)}{3!} (0.0106723) \\
&+ \frac{\frac{1}{3} \left(\frac{1}{3} - 1 \right) \left(\frac{1}{3} - 2 \right) \left(\frac{1}{3} - 3 \right)}{4!} (0.0003548) \\
&= 0.7651977 + \left(\frac{1}{3} \right) (-0.1451117) + \frac{1}{3} \left(-\frac{2}{3} \right) \left(\frac{1}{2} \right) (-0.0195721) \\
&+ \frac{1}{3} \left(-\frac{2}{3} \right) \left(-\frac{5}{3} \right) \left(\frac{1}{6} \right) (0.0106723) \\
&+ \frac{1}{3} \left(-\frac{2}{3} \right) \left(-\frac{5}{3} \right) \left(-\frac{8}{3} \right) \left(\frac{1}{24} \right) (0.0003548) \\
&= 0.719646
\end{aligned}$$

Newton Backward-Difference Formula

❖ Suppose the interpolating nodes are reordered from last to first as $x_n, x_{n-1}, \dots, x_0$.

✓ The we can write the interpolating polynomial as

$$P_n(x) = f[x_n] + f[x_n, x_{n-1}](x - x_n) + f[x_n, x_{n-1}, x_{n-2}](x - x_n)(x - x_{n-1}) \\ + \dots + f[x_n, x_{n-1}, \dots, x_0](x - x_n)(x - x_{n-1}) \dots (x - x_0)$$

❖ If in addition, the nodes are equally spaced with $x = x_n + sh$

✓ It follows that

$$x = x_n + sh$$

$$x - x_{n-1} = x_n + sh - (x_n - h) \quad x_n - h = x_{n-1}$$

$$x - x_{n-1} = sh + h$$

$$x = x_{n-1} + (s + 1)h$$

Newton Backward-Difference Formula

$$x = x_{n-1} + (s + 1)h$$

$$x - x_{n-2} = x_{n-1} + (s + 1)h - x_{n-2}$$

$$x - x_{n-2} = x_{n-1} + (s + 1)h - (x_{n-1} - h)$$

$$x - x_{n-2} = (s + 1)h + h$$

$$x = x_{n-2} + (s + 1)h + h$$

$$x = x_{n-2} + (s + 2)h$$

✓ In general, we have

$$x = x_i + (s + n - i)h$$

Newton Backward-Difference Formula

❖ Consequently, with $x - x_i = (s + n - i)h$, we can rewrite

$$P_n(x) = f[x_n] + f[x_n, x_{n-1}](x - x_n) + f[x_n, x_{n-1}, x_{n-2}](x - x_n)(x - x_{n-1}) \\ + \cdots + f[x_n, x_{n-1}, \dots, x_0](x - x_n)(x - x_{n-1}) \cdots (x - x_0)$$

as

$$P_n(x) = f[x_n] + shf[x_n, x_{n-1}] + s(s + 1)h^2f[x_n, x_{n-1}, x_{n-2}] \\ + \cdots + s(s + 1) \cdots (s + n - 1)h^n f[x_n, x_{n-1}, \dots, x_0]$$

Newton Backward-Difference Formula

Definition: For a given sequence $\{p_n\}_{n=0}^{\infty}$, the backward difference ∇p_n (read “nabla p_n ”) is defined by

$$\nabla p_n = p_n - p_{n-1}, \text{ for } n \geq 1$$

Higher powers of the operator ∇ are defined recursively by $\nabla^k p_n = \nabla(\nabla^{k-1} p_n)$, for $k \geq 2$

Newton Backward-Difference Formula

Definition: For a given sequence $\{p_n\}_{n=0}^{\infty}$, the backward difference ∇p_n (read “nabla p_n ”) is defined by

$$\nabla p_n = p_n - p_{n-1}, \text{ for } n \geq 1$$

Higher powers of the operator ∇ are defined recursively by $\nabla^k p_n = \nabla(\nabla^{k-1} p_n)$, for $k \geq 2$

✓ The definition implies that

$$f[x_n, x_{n-1}, \dots, x_{n-k}] = \frac{1}{k! h^k} \nabla^k f(x_n)$$

Newton Backward-Difference Formula

❖ Using the backward difference notation $\nabla p_n = p_n - p_{n-1}$, observe:

$$f[x_n, x_{n-1}] = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} = \frac{1}{h} (f(x_n) - f(x_{n-1})) = \frac{1}{h} \nabla f(x_n)$$

$$\begin{aligned} f[x_n, x_{n-1}, x_{n-2}] &= \frac{\left[\frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}} - \frac{f(x_{n-1}) - f(x_{n-2})}{x_{n-1} - x_{n-2}} \right]}{x_n - x_{n-2}} \\ &= \frac{1}{2h} \left(\frac{\nabla f(x_n)}{h} - \frac{\nabla f(x_{n-1})}{h} \right) = \frac{1}{2h^2} \nabla^2 f(x_n) \end{aligned}$$

$$f[x_n, x_{n-1}, \dots, x_{n-k}] = \frac{1}{k! h^k} \nabla^k f(x_n)$$

Newton Backward-Difference Formula

❖ Hence, we can rewrite

$$P_n(x_n + sh) = f[x_n] + shf[x_n, x_{n-1}] + s(s+1)h^2f[x_n, x_{n-1}, x_{n-2}] \\ + \cdots + s(s+1) \cdots (s+n-1)h^n f[x_n, x_{n-1}, \dots, x_0]$$

as

$$P_n(x) = f[x_n] + s\nabla f(x_n) + \frac{s(s+1)}{2}\nabla^2 f(x_n) \\ + \cdots + \frac{s(s+1) \cdots (s+n-1)}{n!}\nabla^n f(x_n)$$

Newton Backward-Difference Formula

❖ If we use

$$\binom{-s}{k} = \frac{-s(-s-1)\cdots(-s-k+1)}{k!} = (-1)^k \frac{s(s+1)\cdots(s+k-1)}{k!}$$

Then we have

$$\begin{aligned} P_n(x_n + sh) = & f[x_n] + (-1)^1 \binom{-s}{1} \nabla f(x_n) + (-1)^2 \binom{-s}{2} \nabla^2 f(x_n) \\ & + \cdots + (-1)^n \binom{-s}{n} \nabla^n f(x_n) \end{aligned}$$

Newton Backward-Difference Formula

❖ We now have the Newton backward-difference formula

$$P_n(x_n + sh) = f[x_n] + \sum_{k=1}^n (-1)^k \binom{-s}{k} \nabla^k f(x_n)$$

wherein

$$\binom{-s}{k} = (-1)^k \frac{s(s+1) \cdots (s+k-1)}{k!}$$

Newton Backward-Difference Formula

Example: Consider the following divided difference table we previously obtained.

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
1	1.3	0.6200860	-0.4837057			
2	1.6	0.4554022	-0.5489460	-0.1087339	0.0658784	
3	1.9	0.2818186	-0.5786120	-0.0494433	0.0680685	0.0018251
4	2.2	0.1103623	-0.5715210	0.0118183		

We will interpolate f at $x = 2$ using the given five points.

Solution:

Given:

$$x = 2 = 2.2 - 0.2$$

$$h = 1.3 - 1.0 = 0.3$$

so $s = \left(-\frac{2}{3}\right)$, that is

$$x = 2 = 2.2 + \left(-\frac{2}{3}\right)(0.3)$$

Using Newton's Divided difference formula:

$$P_n(x) = f[x_n] + shf[x_n, x_{n-1}] + s(s+1)h^2f[x_n, x_{n-1}, x_{n-2}] + \dots + s(s+1) \dots (s+n-1)h^n f[x_n, x_{n-1}, \dots, x_0]$$

Using Newton's Divided difference formula:

$$P_4\left(2.2 + \left(-\frac{2}{3}\right)(0.3)\right) = 0.2238754$$

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
1	1.3	0.6200860	-0.4837057			
2	1.6	0.4554022	-0.5489460	-0.1087339	0.0658784	
3	1.9	0.2818186	-0.5786120	-0.0494433	0.0680685	0.0018251
4	2.2	0.1103623	-0.5715210	0.0118183		

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
			-0.4837057			
1	1.3	0.6200860		-0.1087339		
			-0.5489460		0.0658784	
2	1.6	0.4554022		-0.0494433		0.0018251
			-0.5786120		0.0680685	
3	1.9	0.2818186		0.0118183		
			-0.5715210			
4	2.2	0.1103623				
		$f[x_4] = 0.1103623$				

Using Newton's Divided difference formula:

$$P_4(x)$$

$$\begin{aligned}
 &= f[x_4] + shf[x_4, x_3] + s(s+1)h^2f[x_4, x_3, x_2] + s(s+1)(s+2)h^3f[x_4, x_3, x_2, x_1] \\
 &\quad + s(s+1)(s+2)(s+3)h^4f[x_4, x_3, x_2, x_1, x_0]
 \end{aligned}$$

$$f[x_4, x_3] = -0.5715210$$

$$f[x_4, x_3, x_2] = 0.0118183$$

$$f[x_4, x_3, x_2, x_1] = 0.0680685$$

$$f[x_4, x_3, x_2, x_1, x_0] = 0.0018251$$

Using Newton's Divided difference formula:

$$P_4(x) = f[x_4] + shf[x_4, x_3] + s(s+1)h^2f[x_4, x_3, x_2] \\ + s(s+1)(s+2)h^3f[x_4, x_3, x_2, x_1] \\ + s(s+1)(s+2)(s+3)h^4f[x_4, x_3, x_2, x_1, x_0]$$

$f[x_4] = 0.1103623$
 $f[x_4, x_3] = -0.5715210$

$$= 0.1103623 + \left(-\frac{2}{3}\right)(0.3)(-0.5715210) \\ + \left(-\frac{2}{3}\right)\left(\left(-\frac{2}{3}\right) + 1\right)(0.3)^2(0.0118183)$$

$f[x_4, x_3, x_2] = 0.0118183$
 $f[x_4, x_3, x_2, x_1] = 0.0680685$
 $f[x_4, x_3, x_2, x_1, x_0] = 0.0018251$

$$+ \left(-\frac{2}{3}\right)\left(\left(-\frac{2}{3}\right) + 1\right)\left(\left(-\frac{2}{3}\right) + 2\right)(0.3)^3(0.0680685) \\ + \left(-\frac{2}{3}\right)\left(\left(-\frac{2}{3}\right) + 1\right)\left(\left(-\frac{2}{3}\right) + 2\right)\left(\left(-\frac{2}{3}\right) + 3\right)(0.3)^4(0.0018251)$$

$s = -\frac{2}{3}, h = 0.3$

$$= 0.2238754$$

Solution:

Given:

$$x = 2 = 2.2 - 0.2$$

$$h = 1.3 - 1.0 = 0.3$$

so $s = \frac{2}{3}$, that is

$$x = 2 = 2.2 + \left(-\frac{2}{3}\right)(0.3)$$

Using Newton Backward difference formula:

$$P_n(x) = f[x_n] + \sum_{k=1}^n (-1)^k \binom{-s}{k} \nabla^k f(x_n)$$

$$P_n(x) = f[x_n] + (-1)^1 \binom{-s}{1} \nabla f(x_n) + (-1)^2 \binom{-s}{2} \nabla^2 f(x_n) + (-1)^3 \binom{-s}{3} \nabla^3 f(x_n) + (-1)^4 \binom{-s}{4} \nabla^4 f(x_n)$$

i	x_i	$f[x_i]$	$f[x_{i-1}, x_i]$	$f[x_{i-2}, x_{i-1}, x_i]$	$f[x_{i-3}, \dots, x_i]$	$f[x_{i-4}, \dots, x_i]$
0	1.0	0.7651977				
1	1.3	0.6200860	-0.4837057			
2	1.6	0.4554022	-0.5489460	-0.1087339	0.0658784	
3	1.9	0.2818186	-0.5786120	-0.0494433	0.0680685	0.0018251
4	2.2	0.1103623	-0.5715210	0.0118183		

$$\nabla f(x_n) = f(x_n) - f(x_{n-1})$$

$$\nabla p_n = p_n - p_{n-1}$$

$$\begin{aligned}\nabla^2 f(x_n) &= \nabla f(x_n) - \nabla f(x_{n-1}) \\ &= f(x_n) - f(x_{n-1}) - (f(x_{n-1}) - f(x_{n-2})) \\ &= f(x_n) - 2f(x_{n-1}) + f(x_{n-2})\end{aligned}$$

$$\begin{aligned}\nabla^3 f(x_n) &= \nabla f(x_n) - 2\nabla f(x_{n-1}) + \nabla f(x_{n-2}) \\ &= f(x_n) - f(x_{n-1}) - 2(f(x_{n-1}) - f(x_{n-2})) \\ &\quad + (f(x_{n-2}) - f(x_{n-3})) \\ &= f(x_n) - 3f(x_{n-1}) + 3f(x_{n-2}) - f(x_{n-3})\end{aligned}$$

$$\begin{aligned}\nabla^4 f(x_n) &= \nabla f(x_n) - 3\nabla f(x_{n-1}) + 3\nabla f(x_{n-2}) - \nabla f(x_{n-3}) \\ &= f(x_n) - 4f(x_{n-1}) + 6f(x_{n-2}) - 4f(x_{n-3}) + f(x_{n-4})\end{aligned}$$

$$\begin{aligned}\nabla f(x_4) &= f(x_4) - f(x_3) \\ &= 0.1103623 - 0.2818186 \\ &= -0.1714563\end{aligned}$$

$$\begin{aligned}f(x_0) &= 0.7651977 \\ f(x_1) &= 0.6200860 \\ f(x_2) &= 0.4554022 \\ f(x_3) &= 0.2818186 \\ f(x_4) &= 0.1103623\end{aligned}$$

$$\begin{aligned}\nabla^2 f(x_4) &= f(x_4) - 2f(x_3) + f(x_2) \\ &= 0.1103623 - 2(0.2818186) + 0.4554022 \\ &= 0.0021273\end{aligned}$$

$$\begin{aligned}\nabla^3 f(x_4) &= f(x_4) - 3f(x_3) + 3f(x_2) - f(x_1) \\ &= 0.1103623 - 3(0.2818186) + 3(0.4554022) - 0.6200860 \\ &= 0.0110271\end{aligned}$$

$$\begin{aligned}\nabla^4 f(x_4) &= f(x_4) - 4f(x_3) + 6f(x_2) - 4f(x_1) + f(x_0) \\ &= 0.1103623 - 4(0.2818186) + 6(0.4554022) - 4(0.6200860) \\ &\quad + 0.7651977 \\ &= 0.0003548\end{aligned}$$

$$P_n(x) = f[x_n] + (-1)^1 \binom{-s}{1} \nabla f(x_n) + (-1)^2 \binom{-s}{2} \nabla^2 f(x_n) + (-1)^3 \binom{-s}{3} \nabla^3 f(x_n) + (-1)^4 \binom{-s}{4} \nabla^4 f(x_n)$$

$$= 0.1103623$$

$$+ (-1)^1 \binom{2/3}{1} (-0.1714563)$$

$$+ (-1)^2 \binom{2/3}{2} (0.0021273)$$

$$+ (-1)^3 \binom{2/3}{3} (0.0110271)$$

$$+ (-1)^4 \binom{2/3}{4} (0.0003548)$$

$$\nabla f(x_4) = -0.1714563$$

$$\nabla^2 f(x_4) = 0.0021273$$

$$\nabla^3 f(x_4) = 0.0110271$$

$$\nabla^4 f(x_4) = 0.0003548$$

$$s = -\frac{2}{3}, h = 0.3$$

$$f[x_4] = 0.1103623$$

$$P_n(x) = f[x_n] + (-1)^1 \binom{-s}{1} \nabla f(x_n) + (-1)^2 \binom{-s}{2} \nabla^2 f(x_n) + \dots + \frac{s(s+1) \cdots (s+n-1)}{n!} \nabla^n f(x_n)$$

$$= 0.1103623 + (-1)^1 \binom{2/3}{1} (-0.1714563) + (-1)^2 \binom{2/3}{2} (0.0021273) \\ + (-1)^3 \binom{2/3}{3} (0.0110271) + (-1)^4 \binom{2/3}{4} (0.0003548)$$

$$= 0.1103623 - \binom{2/3}{1} (-0.1714563) + \binom{2/3}{2} (0.0021273) \\ - \binom{2/3}{3} (0.0110271) + \binom{2/3}{4} (0.0003548)$$

$$= 0.1103623 - \left(\frac{2}{3}\right) (-0.1714563) + \left(\frac{2}{3}\right) \left(\frac{2}{3} - 1\right) \left(\frac{1}{2}\right) (0.0021273) \\ - \left(\frac{2}{3}\right) \left(\frac{2}{3} - 1\right) \left(\frac{2}{3} - 2\right) \left(\frac{1}{6}\right) (0.0110271) \\ + \left(\frac{2}{3}\right) \left(\frac{2}{3} - 1\right) \left(\frac{2}{3} - 2\right) \left(\frac{2}{3} - 3\right) \left(\frac{1}{24}\right) (0.0003548)$$

$$= 0.2238754$$

Centered-Difference Formula

- ❖ The Newton forward- and backward-difference formulas are not appropriate for approximating $f(x)$ when x lies near the center of the table because neither will permit the highest-order difference to have x_0 close to x .

i	x_i
0	1.0
1	1.3
2	1.6
3	1.9
4	2.2

- ✓ The formulas rely on factors $(x - x_0), (x - x_1), (x - x_2), \dots$
 - ✓ If x is far from x_0 (forward) or far from x_n (backward), these terms get large.
 - ✓ This makes higher-order differences less stable and more prone to numerical error
- ❖ We have centered-difference formulas which is advantageous for cases when x lies near the center of the table

Centered-Difference Formula

❖ We will only consider one centered-difference formula: **Stirling's Method**

- ✓ We choose x_0 near the point being approximated and label the nodes directly below x_0 as $x_1, x_2, \dots$ and those directly above as $x_{-1}, x_{-2}, \dots$.

x	$f(x)$	First divided differences	Second divided differences	Third divided differences	Fourth divided differences
x_{-2}	$f[x_{-2}]$				
		$f[x_{-2}, x_{-1}]$			
x_{-1}	$f[x_{-1}]$		$f[x_{-2}, x_{-1}, x_0]$		
		$\underline{f[x_{-1}, x_0]}$		$\underline{f[x_{-2}, x_{-1}, x_0, x_1]}$	
x_0	$\underline{f[x_0]}$		$\underline{f[x_{-1}, x_0, x_1]}$		$\underline{f[x_{-2}, x_{-1}, x_0, x_1, x_2]}$
		$\underline{f[x_0, x_1]}$		$\underline{f[x_{-1}, x_0, x_1, x_2]}$	
x_1	$f[x_1]$		$f[x_0, x_1, x_2]$		
		$f[x_1, x_2]$			
x_2	$f[x_2]$				

Centered-Difference Formula

Stirling's Formula

$$\begin{aligned} P_n(x) = P_{2m+1}(x) = & f[x_0] + \frac{sh}{2} (f[x_{-1}, x_0] + f[x_0, x_1]) + \frac{s^2 h^2}{2} f[x_{-1}, x_0, x_1] \\ & + \frac{s(s^2 - 1)h^3}{2} (f[x_{-2}, x_{-1}, x_0, x_1] + f[x_{-1}, x_0, x_1, x_2]) \\ & \quad + s^2(s^2 - 1)h^4 f[x_{-2}, x_{-1}, x_0, x_1, x_2] \\ & \dots \\ & + s^2(s^2 - 1)(s^2 - 4) \dots (s^2 - (m - 1)^2) h^{2m} f[x_{-m}, x_{-(m-1)}, \dots, x_m] \\ & + \frac{s(s^2 - 1)(s^2 - 4) \dots (s^2 - m^2) h^{2m+1}}{2} (f[x_{-m-1}, \dots, x_m] + f[x_{-m}, \dots, x_{m+1}]) \end{aligned}$$

Centered-Difference Formula

Stirling's Formula

$$\begin{aligned} P_n(x) = P_{2m+1}(x) = & f[x_0] + \frac{sh}{2} (f[x_{-1}, x_0] + f[x_0, x_1]) + \frac{s^2 h^2}{2} f[x_{-1}, x_0, x_1] \\ & + \frac{s(s^2 - 1)h^3}{2} (f[x_{-2}, x_{-1}, x_0, x_1] + f[x_{-1}, x_0, x_1, x_2]) + s^2(s^2 - 1)h^4 f[x_{-2}, x_{-1}, x_0, x_1, x_2] \\ & + \frac{s(s^2 - 1)(s^2 - 4)h^5}{2} (f[x_{-3}, x_{-2}, x_{-1}, x_0, x_1, x_2] + f[x_{-2}, x_{-1}, x_0, x_1, x_2, x_3]) \\ & + s^2(s^2 - 1)(s^2 - 4)h^6 f[x_{-3}, x_{-2}, x_{-1}, x_0, x_1, x_2, x_3] \\ & \dots \\ & + s^2(s^2 - 1)(s^2 - 4) \dots (s^2 - (m - 1)^2)h^{2m} f[x_{-m}, x_{-(m-1)}, \dots, x_m] \\ & + \frac{s(s^2 - 1)(s^2 - 4) \dots (s^2 - m^2)h^{2m+1}}{2} (f[x_{-m-1}, \dots, x_m] + f[x_{-m}, \dots, x_{m+1}]) \end{aligned}$$

Centered-Difference Formula

Example: Consider the following divided difference table we previously obtained.

x	$f(x)$	First divided differences	Second divided differences	Third divided differences	Fourth divided differences
1.0	0.7651977				
		-0.4837057			
1.3	0.6200860		-0.1087339		
		<u>-0.5489460</u>		<u>0.0658784</u>	
1.6	<u>0.4554022</u>		<u>-0.0494433</u>		<u>0.0018251</u>
		<u>-0.5786120</u>		<u>0.0680685</u>	
1.9	0.2818186		0.0118183		
		-0.5715210			
2.2	0.1103623				

We use Stirling's formula to approximate $f(1.5)$ with $x_0 = 1.6$.

Centered-Difference Formula

Solution: To apply Stirling's formula we use the underlined entries on the previous table.

The formula, with $h = 0.3$, $x_0 = 1.6$, and $s = -\frac{1}{3}$, becomes

$$\begin{aligned} f(1.5) &\approx P_4 \left(1.6 + \left(-\frac{1}{3} \right) (0.3) \right) \\ &= 0.4554022 + \left(-\frac{1}{3} \right) \left(\frac{0.3}{2} \right) ((-0.5489460) + (-0.5786120)) \\ &\quad + \left(-\frac{1}{3} \right)^2 (0.3)^2 (-0.0494433) \\ &\quad + \frac{1}{2} \left(-\frac{1}{3} \right) \left(\left(-\frac{1}{3} \right)^2 - 1 \right) (0.3)^3 (0.0658784 + 0.0680685) \\ &\quad + \left(-\frac{1}{3} \right)^2 \left(\left(-\frac{1}{3} \right)^2 - 1 \right) (0.3)^4 (0.0018251) = 0.5118200. \end{aligned}$$



Osculating Polynomial

Definition: Let $x_0, x_1, x_2, \dots, x_n$ be $n + 1$ distinct numbers in $[a, b]$ for $i = 0, 1, \dots, n$, let m_i be a nonnegative integer. Suppose $f \in C^m[a, b]$ where $m = \max_{0 \leq i \leq n} m_i$.

The **osculating polynomial** approximating f is the polynomial $P(x)$ of least degree (degree of $P(x) \leq n + \sum_{i=0}^n m_i$) that has the same values as the function f and all its derivatives of order less than or equal to m_i at each x_i , i.e.,

$$\frac{d^k P(x_i)}{dx^k} = \frac{d^k f(x_i)}{dx^k}$$

for each $i = 0, 1, \dots, n$ and $k = 0, 1, 2, \dots, m_i$.

- ✓ When $n = 0$, the osculating polynomial approximating f is the m_0 th **Taylor polynomial** for f at x_0 .

Osculating Polynomial(Taylor Polynomial)

Theorem: [Taylor's Theorem]

Suppose $f \in C^m[a, b]$, that $f^{(n+1)}$ exists on $[a, b]$, and $x_0 \in [a, b]$. For every $x \in [a, b]$, there exist a number $\xi(x)$ between x_0 and x with

$$f(x) = P_n(x) + R_n(x)$$

where

$$P_n(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \cdots + \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

$$= \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

$$R_n(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!} (x - x_0)^{n+1}$$

If $n=2$,

$$P_2(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2$$

$$R_2(x) = \frac{f^{(2+1)}(\xi(x))}{(2+1)!} (x - x_0)^{2+1}$$

Osculating Polynomial(Taylor Polynomial)

Example: Let $f(x) = \cos x$ and $x_0 = 0$. Determine
a.) the second Taylor polynomial for f about x_0 ; and
b.) the third Taylor polynomial for f about x_0

Solution Since $f \in C^\infty(\mathbb{R})$, Taylor's Theorem can be applied for any $n \geq 0$. Also,

$$f'(x) = -\sin x, \quad f''(x) = -\cos x, \quad f'''(x) = \sin x, \quad \text{and} \quad f^{(4)}(x) = \cos x,$$

so

$$f(0) = 1, \quad f'(0) = 0, \quad f''(0) = -1, \quad \text{and} \quad f'''(0) = 0.$$

(a) For $n = 2$ and $x_0 = 0$, we have

$$\begin{aligned} \cos x &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(\xi(x))}{3!}x^3 \\ &= 1 - \frac{1}{2}x^2 + \frac{1}{6}x^3 \sin \xi(x), \end{aligned}$$

Osculating Polynomial(Taylor Polynomial)

- ❖ The infinite series obtained by taking the limit of $P_n(x)$ as $n \rightarrow \infty$ is called the Taylor series for f about x_0 .
- ❖ In the case $x_0 = 0$, the Taylor polynomial is often called a **Maclaurin polynomial**. In this case, the Taylor series is called a **Maclaurin series**.

Example: Determine the Taylor series for $f(x) = \sin x$ with $x_0 = 0$. (Or Taylor series centered at $x = 0$)

Answer:

$$\sin x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{(2n+1)!}$$

Osculating Polynomial

Definition: Let $x_0, x_1, x_2, \dots, x_n$ be $n + 1$ distinct numbers in $[a, b]$ for $i = 0, 1, \dots, n$, let m_i be a nonnegative integer. Suppose $f \in C^m[a, b]$ where $m = \max_{0 \leq i \leq n} m_i$.

The **osculating polynomial** approximating f is the polynomial $P(x)$ of least degree (degree of $P(x) \leq n + \sum_{i=0}^n m_i$) that has the same values as the function f and all its derivatives of order less than or equal to m_i at each x_i , i.e.,

$$\frac{d^k P(x_i)}{dx^k} = \frac{d^k f(x_i)}{dx^k}$$

for each $i = 0, 1, \dots, n$ and $k = 0, 1, 2, \dots, m_i$.

- ✓ When $m_i = 0$ for each i , the osculating polynomial is the n th **Lagrange polynomial** interpolating f on $x_0, x_1, x_2, \dots, x_n$.

Osculating Polynomial

Definition: Let $x_0, x_1, x_2, \dots, x_n$ be $n + 1$ distinct numbers in $[a, b]$ for $i = 0, 1, \dots, n$, let m_i be a nonnegative integer. Suppose $f \in C^m[a, b]$ where $m = \max_{0 \leq i \leq n} m_i$.

The **osculating polynomial** approximating f is the polynomial $P(x)$ of least degree (degree of $P(x) \leq n + \sum_{i=0}^n m_i$) that has the same values as the function f and all its derivatives of order less than or equal to m_i at each x_i , i.e.,

$$\frac{d^k P(x_i)}{dx^k} = \frac{d^k f(x_i)}{dx^k}$$

for each $i = 0, 1, \dots, n$ and $k = 0, 1, 2, \dots, m_i$.

- ✓ When $m_i = 1$ for each $i = 0, 1, \dots, n$, the osculating polynomial is the **Hermite polynomial** interpolating f on $x_0, x_1, x_2, \dots, x_n$.

Hermite Interpolation

- ❖ The case when $m_i = 1$, for each $i = 0, 1, \dots, n$, gives the Hermite polynomials.
- ❖ For a given function f , these polynomials agree with f at $x_0, x_1, \dots, x_n$.
- ❖ Since their first derivatives agree with those of f , they have the same “shape” as the function at $(x_i, f(x_i))$ in the sense that the tangent lines to the polynomial and the function agree.

Hermite Interpolation

Theorem: If $f \in C^1[a, b]$ and $x_0, x_1, x_2, \dots, x_n \in [a, b]$ are distinct, the unique polynomial of least degree agreeing with f and f' at $x_0, x_1, x_2, \dots, x_n$ is the Hermite polynomial of degree at most $2n + 1$ given by

$$H_{2n+1}(x) = \sum_{j=0}^n f(x_j) H_{n,j}(x) + \sum_{j=0}^n f'(x_j) \hat{H}_{n,j}(x)$$

$H_{2,0}$
 $\hat{H}_{2,0}$

where, for $L_{n,j}(x)$ denoting the j th Lagrange coefficient polynomial of degree n , we have

$$H_{n,j}(x) = [1 - 2(x - x_j)L'_{n,j}(x_j)]L_{n,j}^2(x)$$

$$\hat{H}_{n,j}(x) = (x - x_j)L_{n,j}^2(x).$$

Moreover, if $f \in C^{2n+2}[a, b]$, then

$$f(x) = H_{2n+1}(x) + \frac{(x - x_0)^2 \dots (x - x_n)^2}{(2n + 1)!} f^{(2n+2)}(\xi(x))$$

for some (generally unknown) $\xi(x)$ in the interval (a, b) .

Hermite Interpolation

Example: Use the Hermite polynomial that agrees with the data listed on the right to find an approximation of $f(1.5)$.

Solution: Given 3 points, we have $n = 2$. We need $L_{2,j}(x)$ where $j = 0, 1, 2$.

$$L_{2,0}(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)}$$

$$L_{2,1}(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)}$$

$$L_{2,2}(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)}$$

k	x_k	$f(x_k)$	$f'(x_k)$
0	1.3	0.6200860	-0.5220232
1	1.6	0.4554022	-0.5698959
2	1.9	0.2818186	-0.5811571

$$= \frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9}$$

$$= -\frac{100}{9}x^2 + \frac{320}{9}x - \frac{247}{9}$$

$$= \left(\frac{50}{9}x^2 - \frac{145}{9}x + \frac{104}{9} \right)^2$$

Hermite Interpolation

$$L_{2,0}(x) = \frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9}$$

$$L'_{2,0}(x) = \frac{100}{9}x - \frac{175}{9}$$

$$L_{2,1}(x) = -\frac{100}{9}x^2 + \frac{320}{9}x - \frac{247}{9}$$

$$L'_{2,1}(x) = -\frac{200}{9}x + \frac{320}{9}$$

$$L_{2,2}(x) = \frac{50}{9}x^2 - \frac{145}{9}x + \frac{104}{9}$$

$$L'_{2,2}(x) = \frac{100}{9}x - \frac{145}{9}$$

We need $H_{2,j}(x) = [1 - 2(x - x_j)L'_{2,j}(x_j)]L_{2,j}^2(x)$

$\hat{H}_{2,j}(x) = \underline{(x - x_j)L_{2,j}^2(x)}$ where $j = 0, 1, 2$.

$x_0 = 1.3, x_1 = 1.6$ and $x_2 = 1.9$

Hermite Interpolation

$$H_{2,0}(x) = [1 - 2(x - x_0)L'_{2,0}(x_0)]L_{2,0}^2(x)$$

$$= [1 - 2(x - 1.3)L'_{2,0}(1.3)]L_{2,0}^2(x)$$

$$= [1 - 2(x - 1.3)(-5)] \left(\frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9} \right)^2$$

$$= (10x - 12) \left(\frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9} \right)^2$$

$$\hat{H}_{2,0}(x) = (x - x_0)L_{2,0}^2(x)$$

$$= (x - 1.3) \left(\frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9} \right)^2$$

$$x_0 = 1.3$$
$$L'_{2,0}(x) = \frac{100}{9}x - \frac{175}{9}$$

$$L'_{2,0}(1.3) = \frac{100}{9}(1.3) - \frac{175}{9}$$

$$L_{2,0}^2(x) = \left(\frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9} \right)^2$$

Hermite Interpolation

$$H_{2,0}(x) = [1 - 2(x - x_0)L'_{2,0}(x_0)]$$

$$\hat{H}_{2,0}(x) = (x - x_0)L_{2,0}^2(x)$$

$$H_{2,1}(x) = [1 - 2(x - x_1)L'_{2,1}(x_1)]$$

$$\hat{H}_{2,1}(x) = (x - x_1)L_{2,1}^2(x)$$

$$H_{2,2}(x) = [1 - 2(x - x_2)L'_{2,2}(x_2)]$$

$$\hat{H}_{2,2}(x) = (x - x_2)L_{2,2}^2(x)$$

The polynomials $H_{2,j}(x)$ and $\hat{H}_{2,j}(x)$ are then

$$\begin{aligned} H_{2,0}(x) &= [1 - 2(x - 1.3)(-5)] \left(\frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9} \right)^2 \\ &= (10x - 12) \left(\frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9} \right)^2, \end{aligned}$$

$$H_{2,1}(x) = 1 \cdot \left(\frac{-100}{9}x^2 + \frac{320}{9}x - \frac{247}{9} \right)^2,$$

$$H_{2,2}(x) = 10(2 - x) \left(\frac{50}{9}x^2 - \frac{145}{9}x + \frac{104}{9} \right)^2,$$

$$\hat{H}_{2,0}(x) = (x - 1.3) \left(\frac{50}{9}x^2 - \frac{175}{9}x + \frac{152}{9} \right)^2,$$

$$\hat{H}_{2,1}(x) = (x - 1.6) \left(\frac{-100}{9}x^2 + \frac{320}{9}x - \frac{247}{9} \right)^2,$$

and

$$\hat{H}_{2,2}(x) = (x - 1.9) \left(\frac{50}{9}x^2 - \frac{145}{9}x + \frac{104}{9} \right)^2.$$

Hermite Interpolation

Finally

$$H_5(x) = 0.6200860H_{2,0}(x) + 0.4554022H_{2,1}(x) + 0.2818186H_{2,2}(x) \\ - 0.5220232\hat{H}_{2,0}(x) - 0.5698959\hat{H}_{2,1}(x) - 0.5811571\hat{H}_{2,2}(x)$$

$f'(x_0)$ $f'(x_1)$ $f'(x_2)$

and

$$H_5(1.5) = 0.6200860 \left(\frac{4}{27} \right) + 0.4554022 \left(\frac{64}{81} \right) + 0.2818186 \left(\frac{5}{81} \right) \\ - 0.5220232 \left(\frac{4}{405} \right) - 0.5698959 \left(\frac{-32}{405} \right) - 0.5811571 \left(\frac{-2}{405} \right) \\ = 0.5118277,$$

a result that is accurate to the places listed. ■

Hermite Polynomial Using Divided Difference

❖ Suppose that the distinct numbers $x_0, x_1, \dots, x_n$ are given together with the values of f and f' at these numbers.

❖ Define a new sequence $z_0, z_1, \dots, z_{2n+1}$ by

$$z_{2i} = z_{2i+1} = x_i$$

for each $i = 0, 1, \dots, n$, and construct the divided difference table the uses $z_0, z_1, \dots, z_{2n+1}$.

❖ Since $z_{2i} = z_{2i+1} = x_i$ for each i , we cannot define $f[z_{2i}, z_{2i+1}]$ by the divided difference formula.

We use the entries $f'(x_0), f'(x_1), \dots, f'(x_n)$ for
 $f[z_0, z_1], f[z_2, z_3], \dots, f[z_{2n}, z_{2n+1}]$

$$f[z_{2i}, z_{2i+1}] = f'(z_{2i}) = f'(x_i)$$

Hermite Polynomial Using Divided Difference

z	$f(z)$	First divided differences	Second divided differences
$z_0 = x_0$	$f[z_0] = f(x_0)$		
		$f[z_0, z_1] = f'(x_0)$	
$z_1 = x_0$	$f[z_1] = f(x_0)$		$f[z_0, z_1, z_2] = \frac{f[z_1, z_2] - f[z_0, z_1]}{z_2 - z_0}$
		$f[z_1, z_2] = \frac{f[z_2] - f[z_1]}{z_2 - z_1}$	
$z_2 = x_1$	$f[z_2] = f(x_1)$		$f[z_1, z_2, z_3] = \frac{f[z_2, z_3] - f[z_1, z_2]}{z_3 - z_1}$
		$f[z_2, z_3] = f'(x_1)$	
$z_3 = x_1$	$f[z_3] = f(x_1)$		$f[z_2, z_3, z_4] = \frac{f[z_3, z_4] - f[z_2, z_3]}{z_4 - z_2}$
		$f[z_3, z_4] = \frac{f[z_4] - f[z_3]}{z_4 - z_3}$	
$z_4 = x_2$	$f[z_4] = f(x_2)$		$f[z_3, z_4, z_5] = \frac{f[z_4, z_5] - f[z_3, z_4]}{z_5 - z_3}$
		$f[z_4, z_5] = f'(x_2)$	
$z_5 = x_2$	$f[z_5] = f(x_2)$		

Hermite Polynomial Using Divided Difference

Recall:

$$P_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, x_1, x_2, \dots, x_k](x - x_0)(x - x_1) \dots (x - x_{k-1})$$

The Hermite polynomial

$$P_{2n+1}(x) = f[z_0] + \sum_{k=1}^{2n+1} f[z_0, z_1, z_2, \dots, z_k](x - z_0)(x - z_1) \dots (x - z_{k-1})$$

Hermite Polynomial Using Divided Difference

Example: Use the data given on the right and the divided difference method to determine the Hermite polynomial approximation at $x = 1.5$

k	x_k	$f(x_k)$	$f'(x_k)$
0	1.3	0.6200860	-0.5220232
1	1.6	0.4554022	-0.5698959
2	1.9	0.2818186	-0.5811571

<u>1.3</u>	<u>0.6200860</u>					
		<u>-0.5220232</u>				
<u>1.3</u>	<u>0.6200860</u>		-0.0897427			
		-0.5489460		0.0663657		
<u>1.6</u>	<u>0.4554022</u>		-0.0698330		0.0026663	
		<u>-0.5698959</u>		0.0679655		-0.0027738
<u>1.6</u>	<u>0.4554022</u>		-0.0290537		0.0010020	
		-0.5786120		0.0685667		
<u>1.9</u>	<u>0.2818186</u>		-0.0084837			
		<u>-0.5811571</u>				
<u>1.9</u>	<u>0.2818186</u>					

The underlined entries in the first three columns are the given data.
The remaining entries are generated by the standard divided difference formula.

Hermite Polynomial Using Divided Difference

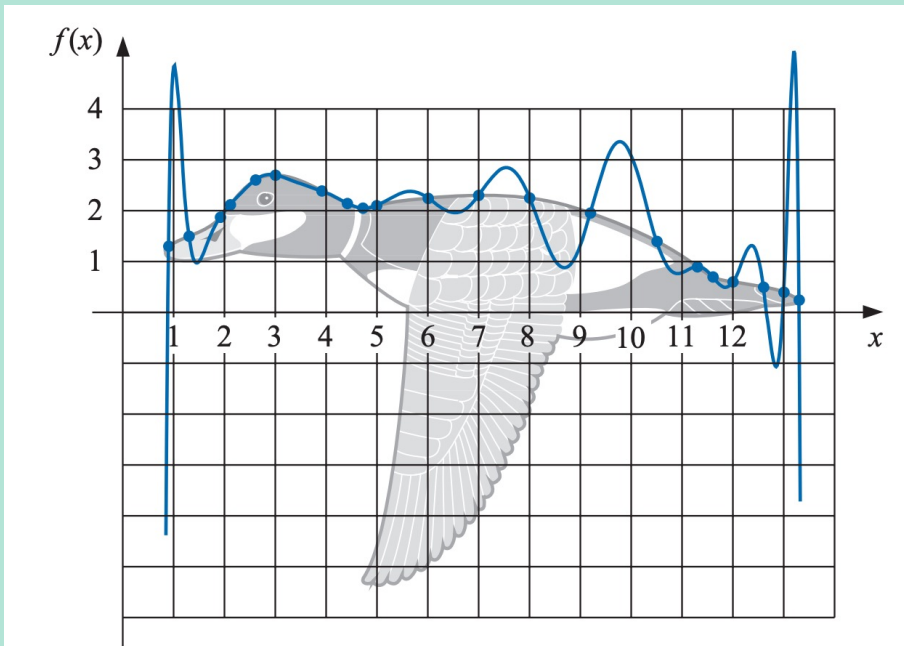
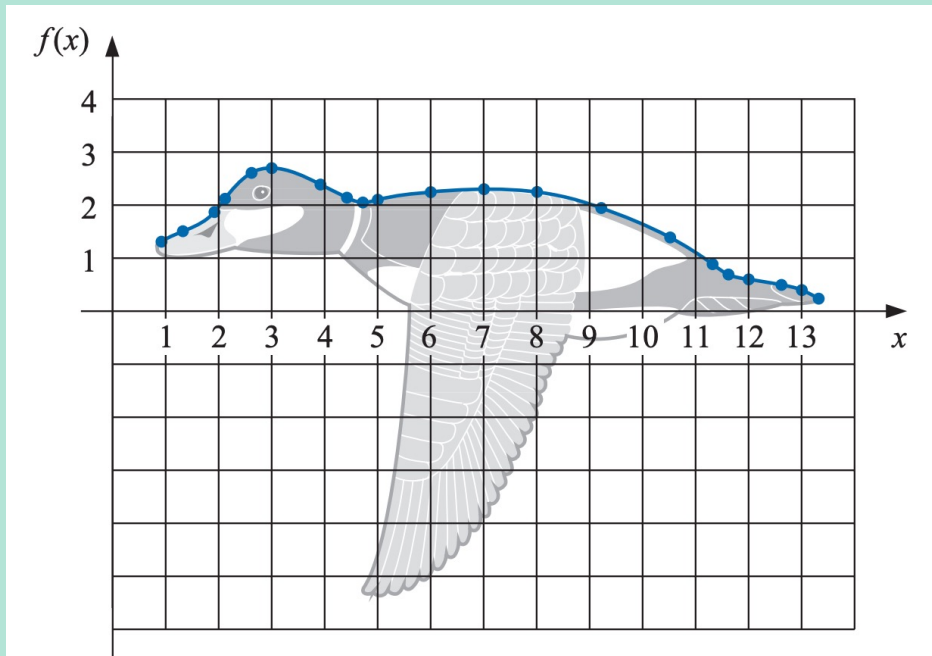
$$P_n(x) = f[x_0] + \sum_{k=1}^n f[x_0, x_1, x_2, \dots, x_k](x - x_0)(x - x_1) \dots (x - x_{k-1})$$

The value of the Hermite polynomial at 1.5 is

$$\begin{aligned} H_5(1.5) &= f[1.3] + f'(1.3)(1.5 - 1.3) + f[1.3, 1.3, 1.6](1.5 - 1.3)^2 \\ &\quad + f[1.3, 1.3, 1.6, 1.6](1.5 - 1.3)^2(1.5 - 1.6) \\ &\quad + f[1.3, 1.3, 1.6, 1.6, 1.9](1.5 - 1.3)^2(1.5 - 1.6)^2 \\ &\quad + f[1.3, 1.3, 1.6, 1.6, 1.9, 1.9](1.5 - 1.3)^2(1.5 - 1.6)^2(1.5 - 1.9) \\ &= 0.6200860 + (-0.5220232)(0.2) + (-0.0897427)(0.2)^2 \\ &\quad + 0.0663657(0.2)^2(-0.1) + 0.0026663(0.2)^2(-0.1)^2 \\ &\quad + (-0.0027738)(0.2)^2(-0.1)^2(-0.4) \\ &= 0.5118277. \end{aligned}$$

Cubic Spline Interpolation

- ❖ The previous methods of interpolation concerned the approximation of arbitrary functions on closed intervals using a single polynomial.
- ❖ However, high-degree polynomials can oscillate erratically, that is, a minor fluctuation over a small portion of the interval can induce large fluctuations over the entire range.

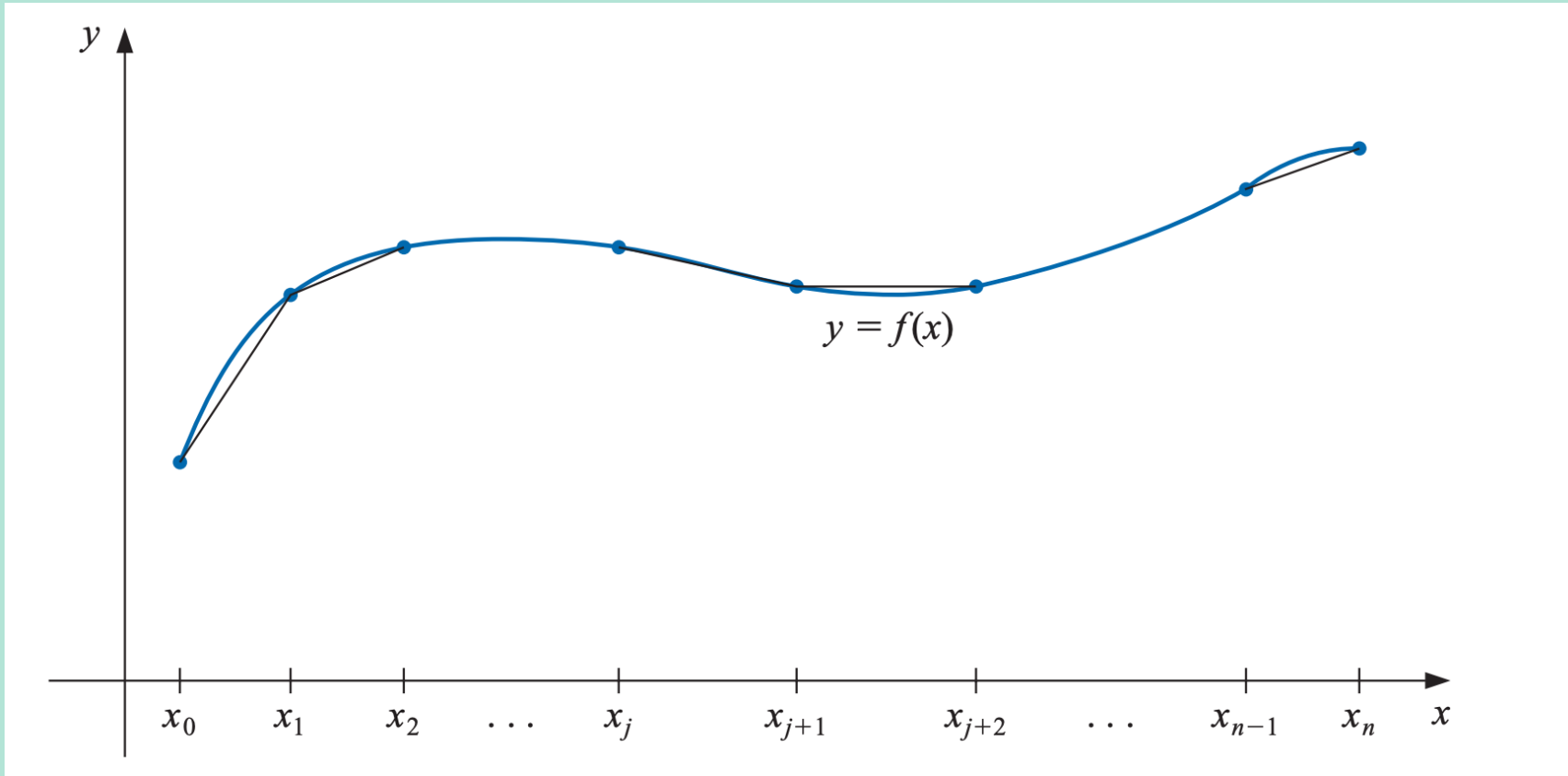


Cubic Spline Interpolation

- ❖ An alternative approach to avoid large number of fluctuations arising from the behavior of higher degree polynomials and minor fluctuation of original data is what we refer to as **piecewise-polynomial approximation**.
- ✓ An alternative approach divides the approximation interval into a collection of subintervals and construct a (generally) different approximating polynomial on each subinterval

Cubic Spline Interpolation

- ❖ The simplest piecewise-polynomial approximation is piecewise-linear interpolation, which consists of joining a set of data points $\{(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))\}$



Cubic Spline Interpolation

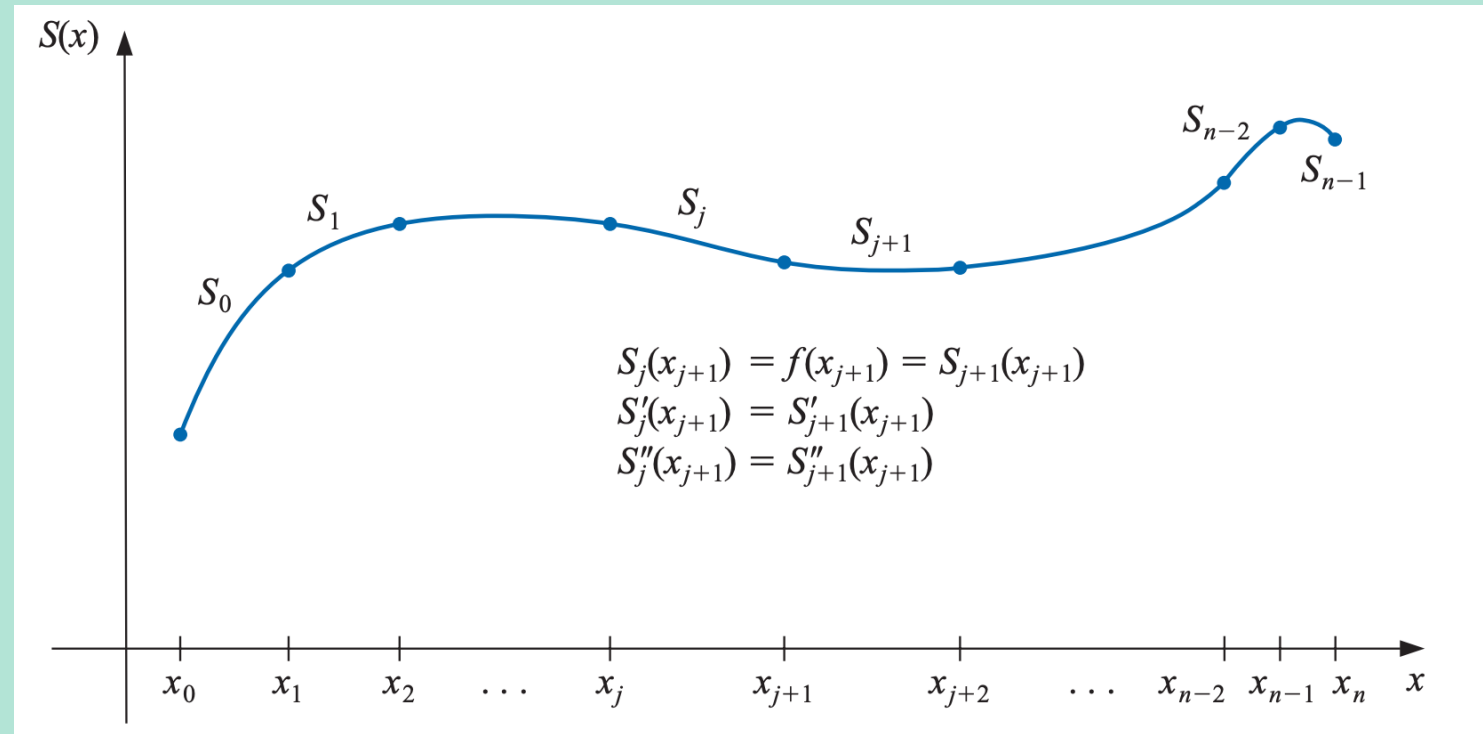
- ❖ The simplest piecewise-polynomial approximation is piecewise-linear interpolation, which consists of joining a set of data points $\{(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))\}$
- ❖ A disadvantage of linear function approximation is that there is likely no differentiability at the endpoints of the subintervals.
 - ✓ Sharp turns or wedge-like graph is not differentiable at the turning point
 - ✓ We want the approximating function to be continuously differentiable
- ❖ An alternative procedure is to use a piecewise polynomial of Hermite type.

Cubic Spline Interpolation

- ❖ To determine the appropriate Hermite cubic polynomial on a given interval is simply a matter of computing $H_3(x)$ for that interval.
- ✓ The Lagrange interpolating polynomials needed to determine H_3 are of first degree, so this can be accomplished without great difficulty.
- ✓ However, we need to know the derivative of the function being approximated, and this is frequently unavailable.
- ❖ We want an approximation using piecewise polynomials that require no specific derivative information, except perhaps at the endpoints of the interval on which the function is being approximated.

Cubic Spline Interpolation

- ❖ The most common piecewise-polynomial approximation uses cubic polynomials between each successive pair of nodes and is called **cubic spline interpolation**.
- ❖ A general cubic polynomial involves four constants, so there is sufficient flexibility in the cubic spline procedure to ensure that the interpolant is not only continuously differentiable on the interval, but also has a continuous second derivative.



Cubic Spline Interpolation

Given a function f defined on $[a, b]$ and a set of nodes $a = x_0 < x_1 < \dots < x_n = b$, a **cubic spline interpolant** S for f is a function that satisfies the following conditions:

- (a) $S(x)$ is a cubic polynomial, denoted $S_j(x)$, on the subinterval $[x_j, x_{j+1}]$ for each $j = 0, 1, \dots, n - 1$;
- (b) $S_j(x_j) = f(x_j)$ and $S_j(x_{j+1}) = f(x_{j+1})$ for each $j = 0, 1, \dots, n - 1$;
- (c) $S_{j+1}(x_{j+1}) = S_j(x_{j+1})$ for each $j = 0, 1, \dots, n - 2$; (*Implied by (b).*)
- (d) $S'_{j+1}(x_{j+1}) = S'_j(x_{j+1})$ for each $j = 0, 1, \dots, n - 2$;
- (e) $S''_{j+1}(x_{j+1}) = S''_j(x_{j+1})$ for each $j = 0, 1, \dots, n - 2$;
- (f) One of the following sets of boundary conditions is satisfied:
 - (i) $S''(x_0) = S''(x_n) = 0$ (**natural (or free) boundary**);
 - (ii) $S'(x_0) = f'(x_0)$ and $S'(x_n) = f'(x_n)$ (**clamped boundary**). ■

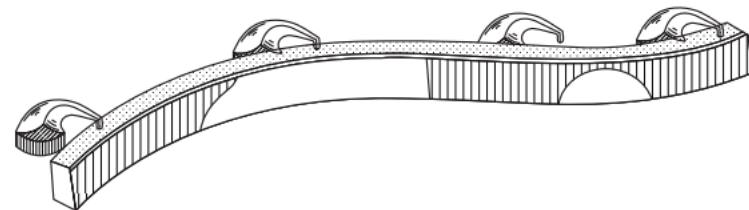
Cubic Spline Interpolation

- ❖ Although cubic splines are defined with other boundary conditions, the conditions given in (f) are sufficient for our purposes.
- ❖ When the free boundary conditions occur, the spline is called a **natural spline**, and its graph approximates the shape that a long flexible rod would assume if forced to go through the data points $\{(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))\}$
- ❖ In general, clamped boundary conditions lead to more accurate approximations because they include more information about the function.
 - ✓ For this type of boundary condition to hold, it is necessary to have either the values of the derivative at the endpoints or an accurate approximation to those values.

Cubic Spline Interpolation

The root of the word “spline” is the same as that of splint. It was originally a small strip of wood that could be used to join two boards. Later the word was used to refer to a long flexible strip, generally of metal, that could be used to draw continuous smooth curves by forcing the strip to pass through specified points and tracing along the curve.

A natural spline has no conditions imposed for the direction at its endpoints, so the curve takes the shape of a straight line after it passes through the interpolation points nearest its endpoints. The name derives from the fact that this is the natural shape a flexible strip assumes if forced to pass through specified interpolation points with no additional constraints.



Cubic Spline Interpolation

Example: Construct a natural cubic spline that passes through the points $(1, 2)$, $(2, 3)$, and $(3, 5)$.